A Survey of On-Policy Distillation for Large Language Models

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Abstract

Knowledge distillation has become a primary mechanism for transferring capabilities from frontier Large Language Models (LLMs) to smaller, deployable students. The dominant paradigm remains *off-policy*: students train on static, teacher-generated data and never encounter their own errors during learning. This train–test mismatch, an instance of *exposure bias* well-studied in imitation learning, causes prediction errors to compound autoregressively at inference time. On-Policy Distillation (OPD) addresses this gap by letting the student generate its own trajectories and receive teacher feedback on these self-generated outputs. Despite rapid growth spanning divergence minimization, reward-guided learning, and self-play, the OPD literature remains fragmented with no unified treatment. This survey provides the first comprehensive overview of OPD for LLMs. We formulate a unified f-divergence framework over student-sampled trajectories and organize the landscape along a primary binary axis: *teacher-guided* methods, which rely on an external teacher through logit access, API queries, or reward signals, and *self-distillation* methods, where the model improves from its own signals without a separate teacher. Each method is further annotated along five complementary dimensions capturing teacher source, signal type, granularity, reinforcement-learning integration, and divergence choice. We analyze representative methods across both categories, examine industrial deployments, and chart open problems including distillation scaling laws, uncertainty-aware feedback, and agent-level distillation.

Awesome List & Paper Collection
https://github.com/nick7nlp/Awesome-LLM-On-Policy-Distillation

1 Introduction

Large Language Models (LLMs) have reshaped natural language processing and, increasingly, artificial intelligence as a whole. Scaling from hundreds of millions of parameters to hundreds of billions, frontier models now demonstrate strong capabilities in reasoning, code generation, multilingual understanding, and instruction following (Yang et al., 2025a; DeepSeek-AI et al., 2025; Gemma Team et al., 2024). Yet the sheer computational cost of training and serving these frontier models remains prohibitive for most deployment scenarios, making the transfer of capabilities from large teachers to compact students not merely desirable but economically necessary. Knowledge distillation (KD), first formalized by Hinton et al. (2015) as training a student network to match the softened output distribution of a teacher, has since grown from a niche compression technique into a central pillar of the modern LLM development pipeline. The release of DeepSeek-R1 (DeepSeek-AI et al., 2025), which successfully transferred complex chain-of-thought reasoning from a 671-billion-parameter mixture-of-experts teacher into dense student models ranging from 1.5 billion to 70 billion parameters, illustrates how distillation now serves as a general-purpose *capability transfer engine* rather than a mere size-reduction tool.

As the role of distillation has expanded, so has awareness of a fundamental bottleneck in its dominant form. Conventional LLM distillation is *off-policy* in nature. The student is trained on a fixed corpus, typically sampled from the data distribution p_{data} or generated in advance by the teacher, and learns to replicate the teacher's token-level probabilities over these pre-collected sequences. At inference time, however, the student must generate autoregressively, conditioning each new token on its own previous (and potentially erroneous) outputs. This discrepancy between the training distribution and the student's own generation distribution creates a *train-test mismatch* that compounds over the length of the sequence. Gudibande et al. (2023) provided a systematic empirical demonstration of this failure mode, showing that off-policy imitation models can mimic a teacher's style without acquiring its factual capabilities, with performance gains largely confined to tasks well-represented in the imitation data. Formally, this phenomenon is an instance of *exposure bias* well-studied in the imitation learning literature (Ross et al., 2011), where a policy trained only on expert-state demonstrations drifts into unfamiliar states at test time and lacks the supervisory signal to recover. For autoregressive LLMs, the problem is particularly severe because errors at early positions propagate through the entire remaining sequence, and the student receives no gradient information about how to behave in these self-induced off-distribution states.

On-Policy Distillation (OPD) offers a principled solution. Rather than training on a static dataset, OPD lets the student sample sequences from its own evolving policy p_θ and then solicits teacher feedback on these self-generated trajectories. This feedback can take diverse forms: full token-level probability distributions in white-box settings (Agarwal et al., 2024; Gu et al., 2024), scalar rewards or pairwise preferences when only black-box teacher access is available (Ye et al., 2025; Jiang et al., 2023), and self-generated contrastive signals in teacher-free frameworks (Chen et al., 2024). The theoretical motivation draws directly from interactive imitation learning. The DAgger algorithm (Ross et al., 2011) established that querying an expert on the learner's own visited states reduces the compounding error from $O(T^2)$ under pure behavior cloning to $O(T)$, where T is the horizon length. OPD instantiates this principle for autoregressive language generation, transforming distillation from one-shot distribution matching into an iterative, self-correcting optimization loop.

This approach has catalyzed rapid methodological innovation along two complementary directions. On the *teacher-guided* side, Generalized Knowledge Distillation (GKD) (Agarwal et al., 2024) introduced on-policy sampling with configurable mixture ratios between student and teacher sequences. MiniLLM (Gu et al., 2024) recast the objective by minimizing Reverse KL divergence to avoid the mode-covering pathology of Forward KL. DistiLLM (Ko et al., 2024) proposed skewed KL objectives that stabilize optimization by mixing student and teacher distributions. RLKD (Xu et al., 2025b) introduced reinforcement learning (RL) with a structure-aware reward model to capture reasoning patterns that distributional matching alone cannot transfer. On the *self-distillation* side, Self-Play Fine-Tuning (SPIN) (Chen et al., 2024) demonstrated that a student can improve by distinguishing its own generations from human references without any teacher model at all, while OPSD (Zhao et al., 2026b) showed that a model can distill from a privileged-information-conditioned version of itself.

Despite this rapid growth, the OPD literature remains fragmented. Methods originating from the knowledge distillation community, the reinforcement learning from human feedback (RLHF) community, and the imitation learning community often address the same underlying problem with different formalisms, different evaluation protocols, and different terminology. Existing surveys of LLM distillation (Xu et al., 2024) predominantly organize the field around the classical compression framing, treating off-policy and on-policy approaches as interchangeable variants rather than as distinct paradigms with different theoretical guarantees and failure modes. To date, no survey has provided a unified mathematical treatment of the on-policy dynamics that connect these seemingly disparate lines of work, nor has any systematically compared the trade-offs among white-box, black-box, and teacher-free instantiations of on-policy learning for LLMs.

This survey fills that gap. We provide the first dedicated and comprehensive treatment of on-policy distillation for Large Language Models, organized around a theoretically motivated taxonomy. Our contributions are as follows.

- **A Unified Theoretical Framework.** We formulate the transition from off-policy to on-policy distillation through the lens of sequential decision-making and f-divergence minimization over student-sampled trajectories. This framework reveals how GKD (Agarwal et al., 2024), MiniLLM (Gu et al., 2024), and DistiLLM (Ko et al., 2024) instantiate the same underlying objective under different divergence choices, argument orderings, target distributions, and sampling mixture coefficients, providing a common analytical language for comparing methods that were previously studied in isolation.
- **A Binary Taxonomy with Multi-Dimensional Annotation.** We organize OPD methods along a primary binary axis, *teacher-guided* distillation (Section 4, organized by five technical challenges) versus *self-distillation* (Section 5, organized by signal source: privileged information, self-play, and external feedback), and annotate each method along five complementary dimensions (teacher source, signal type, granularity, RL integration, and divergence choice). This taxonomy clarifies the design space and reveals underexplored combinations that represent promising research directions.
- **Bridging White-Box and Black-Box Regimes.** We provide a systematic comparison of on-policy methods that require full teacher internals versus those that operate with only sampled outputs, analyzing the theoretical information gap between learning from complete probability distributions and learning from top-k generations through representative methods such as GAD (Ye et al., 2025) and Lion (Jiang et al., 2023).
- **Identification of Overlooked Challenges.** We highlight several issues that have received insufficient attention, including the compute-quality trade-off in online generation, the "echo chamber" effect when querying teachers on low-confidence student states, the need for dynamic divergence adaptation across training stages, and the reconceptualization of distillation as structured capability transfer rather than parameter-level compression.
- **A Roadmap for Future Research.** We chart open problems including distillation scaling laws, uncertainty-aware OPD, curriculum-driven sampling strategies, and agent-level distillation where models must learn from multi-turn, state-dependent interactions with external environments.

The remainder of this survey is organized as follows. Section 2 establishes mathematical preliminaries, formalizing autoregressive generation as a sequential decision problem and deriving the exposure bias that motivates on-policy approaches. Section 3 presents our taxonomy and organization rationale. Section 4 provides an in-depth analysis of teacher-guided on-policy distillation methods, organized around five cumulative technical challenges: objective function design, training dynamics, cross-architecture alignment, information-constrained settings, and supervision beyond distribution matching. The ordering of these subsections is not arbitrary: each addresses a failure mode that emerges after the preceding challenge is resolved, tracing an arc from the foundational divergence choice through practical training stability to the frontier of reward-guided learning. Section 5 covers self-distillation approaches organized by signal source, from privileged-information methods that create an internal teacher-student asymmetry, through self-play methods that exploit temporal diversity, to external-feedback methods that ground self-improvement in verifiable outcomes. Section 6 consolidates theoretical perspectives on OPD success conditions, failure modes, and the on-policy vs. off-policy tradeoff, showing that the empirical patterns across Sections 4 and 5 follow from a small number of unifying principles. Section 7 discusses industrial deployments, system-level optimization, and emerging domains. Section 8 identifies open problems and future directions, organized as a logical agenda from near-term compute bottlenecks to longer-term scientific questions.

2 Background and Preliminaries

Understanding on-policy distillation requires a precise mathematical foundation that connects classical knowledge distillation to the sequential decision-making nature of autoregressive generation. Without this grounding, the exposure bias problem that motivates the entire OPD paradigm remains an informal intuition rather than a quantifiable phenomenon with provable bounds. This section establishes the necessary formalism in four stages. We begin with the classical KD objective and its gradient structure (Section 2.1), extend it to the token-level and sequence-level settings that arise in language modeling (Section 2.2), formalize the distribution mismatch problem that renders off-policy training inherently limited (Section 2.3), and finally introduce the f-divergence family that parameterizes modern OPD objectives (Section 2.4).

Notation. We use lowercase p for token-level conditional distributions (e.g., $p_T(\cdot|x, y_{<t})$, $p_\theta(\cdot|x, y_{<t})$) and uppercase P for sequence-level or trajectory-level distributions (e.g., $P_\mathcal{T}(y|x)$, $P_\theta(y|x)$). In the MDP formalization of exposure bias (Section 2.3), we use π for policies. When discussing individual methods in detail, we follow each paper's notation where helpful (e.g., p_S/p_T for student/teacher in the divergence analysis of Section 4.1, $P_\theta/P_{\text{Teacher}}$ in the reasoning distillation of Section 4.5).

2.1 Knowledge Distillation Fundamentals

The classical knowledge distillation framework, pioneered by Hinton et al. (2015) and comprehensively surveyed by **?**, transfers the implicit "dark knowledge" embedded in the continuous probability distributions of a cumbersome teacher network $\mathcal{T}$ to a compact student network $\mathcal{S}$. In standard classification tasks, and by extension token-level vocabulary selection in language modeling, a neural network produces a vector of pre-softmax logits $z \in \mathbb{R}^{|V|}$, where $|V|$ is the vocabulary size.

To extract the relational structure between classes (for instance, that the token "automobile" is semantically closer to "car" than to "apple"), the logits are smoothed using a temperature scalar $\tau > 0$. The softened probability for a given input context x is

$$p(y|x;\tau) = \frac{\exp(z_y/\tau)}{\sum_{y' \in V} \exp(z_{y'}/\tau)} \tag{1}$$

The standard KD objective seeks to minimize the Kullback-Leibler (KL) divergence between the teacher's softened predictive distribution p_T and the student's corresponding softened distribution p_θ

$$\mathcal{L}_{\text{KD}} = \tau^2 D_{\text{KL}}(p_T(\cdot|x;\tau) \parallel p_\theta(\cdot|x;\tau)) = \tau^2 \sum_{y \in V} p_T(y|x;\tau) \log\left(\frac{p_T(y|x;\tau)}{p_\theta(y|x;\tau)}\right) \tag{2}$$

The τ^2 scaling factor is important because as τ increases, the cross-entropy gradient magnitudes scale down by $1/\tau^2$, so multiplying by τ^2 keeps the distillation loss commensurate with hard-label cross-entropy in a joint objective.

To understand this transfer, we analyze the gradient of the distillation loss with respect to the student's logits $z_i^\mathcal{S}$. Let $p_i^\mathcal{T}$ and $p_i^\mathcal{S}$ denote the temperature-scaled teacher and student probabilities, respectively. The gradient is

$$\frac{\partial \mathcal{L}_{\text{KD}}}{\partial z_i^\mathcal{S}} = \frac{\tau^2}{\tau}(p_i^\mathcal{S} - p_i^\mathcal{T}) = \tau(p_i^\mathcal{S} - p_i^\mathcal{T}) \tag{3}$$

When $\tau \to \infty$, a Taylor expansion $\exp(z_i/\tau) \approx 1 + z_i/\tau$ shows that, assuming zero-meaned logits ($\sum_j z_j = 0$), the probabilities approach

$$p_i \approx \frac{1 + z_i/\tau}{|V| + \sum_j z_j/\tau} = \frac{1}{|V|} + \frac{z_i}{|V|\tau} \tag{4}$$

Substituting back into the gradient yields an equivalence

$$\frac{\partial \mathcal{L}_{\text{KD}}}{\partial z_i^\mathcal{S}} \approx \tau\left(\frac{z_i^\mathcal{S}}{|V|\tau} - \frac{z_i^\mathcal{T}}{|V|\tau}\right) = \frac{1}{|V|}(z_i^\mathcal{S} - z_i^\mathcal{T}) \tag{5}$$

This derivation shows that in the high-temperature limit, classical KD degenerates into minimizing the Mean Squared Error (MSE) between raw logits. This forces the student to replicate the teacher's entire logit structure, transferring both primary predictive modes and nuanced sub-optimal probability masses (the dark knowledge) that underpin generalization. However, when applied naively to autoregressive LLMs, this formulation assumes the input x (the prefix) is drawn from the target data distribution, entirely ignoring the sequential, compounding nature of language generation.

2.2 Token-Level vs. Sequence-Level Distillation

The classical KD framework described above assumes a fixed input x drawn from a static dataset, an assumption that obscures a critical design choice when applied to autoregressive language models. Because each generated token conditions on all preceding tokens, the distillation objective can be factored at two distinct granularities, each with distinct computational and theoretical properties.

Applying Hinton et al. (2015)'s formulation to language modeling typically yields token-level distillation. In an autoregressive setting, a sequence $y = (y_1, y_2, \ldots, y_T)$ is generated conditionally. Let $y_{<t}$ denote the prefix up to step t. The token-level distillation loss is the expected divergence over prefixes from the empirical dataset $\mathcal{D}$.

$$\mathcal{L}_{\text{Token-KD}} = \mathbb{E}_{x,y\sim\mathcal{D}}\left[\sum_{t=1}^{|y|} D_{\text{KL}}(p_T(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t}))\right] \tag{6}$$

While computationally tractable because it relies on static, pre-computed prefixes, token-level KD misaligns with the global generative objective because it optimizes next-token accuracy while assuming an error-free history $y_{<t}$.

Recognizing this limitation, Kim & Rush (2016) proposed Sequence-Level Knowledge Distillation. Instead of factorizing the loss token-by-token over fixed dataset prefixes, sequence-level KD aims to match the joint probability distributions over the entire sequence space $\mathcal{Y}$. The theoretical sequence-level objective minimizes the global KL divergence

$$\mathcal{L}_{\text{Seq-KD}} = D_{\text{KL}}(P_{\mathcal{T}}(y|x) \parallel P_\theta(y|x)) = \sum_{y\in\mathcal{Y}} P_{\mathcal{T}}(y|x)\log\left(\frac{P_{\mathcal{T}}(y|x)}{P_\theta(y|x)}\right) \tag{7}$$

Because the sequence space $\mathcal{Y}$ grows exponentially with length T ($|\mathcal{Y}| = |V|^T$), exactly computing this summation is intractable. Kim & Rush (2016) resolved this pragmatically. Observing that $P_{\mathcal{T}}(y|x)$ is sharply peaked around its mode, they approximated the distribution with a Dirac delta at the beam-search output

$$P_{\mathcal{T}}(y|x) \approx \begin{cases} 1 & \text{if } y = \arg\max_{y'\in\mathcal{Y}} P_{\mathcal{T}}(y'|x) \\ 0 & \text{otherwise} \end{cases} \tag{8}$$

Substituting this approximation into the sequence-level KL divergence simplifies the loss to the standard negative log-likelihood (NLL) of the student, evaluated on the teacher's highest-probability sequence $\hat{y}$

$$\mathcal{L}_{\text{Seq-Approx}} = -\sum_{t=1}^{|\hat{y}|} \log p_\theta(\hat{y}_t|x, \hat{y}_{<t}) \tag{9}$$

This formulation forces the student to model complete trajectories favored by the teacher, implicitly transferring sequence-level structural dependencies. However, Sequence-Level KD remains an *off-policy* algorithm. The student still trains via teacher-forcing on static trajectories (pseudo-labels generated offline by the teacher). While the target $\hat{y}$ originates from a model rather than a human, the student never conditions predictions on its own prefixes during training. Consequently, Sequence-Level KD remains susceptible to the train-test mismatch it sought to alleviate, motivating true on-policy algorithms.

2.3 The Distribution Mismatch Problem and Exposure Bias

Both token-level and sequence-level KD as described above share a common limitation that neither the granularity of the loss function nor the source of the target sequence addresses. Regardless of whether the student matches the teacher token-by-token or trajectory-by-trajectory, training always conditions on prefixes drawn from a fixed dataset, never on the student's own imperfect outputs. This section formalizes the resulting failure mode and quantifies its severity.

The core problem necessitating on-policy distillation is *exposure bias*, a direct manifestation of covariate shift in sequential decision-making. We cast autoregressive generation as a Markov Decision Process (MDP). The state space $\mathcal{S}$ comprises all token prefixes $s_t = (x, y_{<t})$, the action space $\mathcal{A}$ is the vocabulary V, and transitions are deterministic: action y_t in state s_t yields $s_{t+1} = (x, y_{<t}, y_t)$.

During off-policy distillation (teacher-forcing), the student encounters states governed by the data distribution p_{data}. Defining the state visitation distribution under the dataset policy as $d_{\mathcal{D}}(s)$, the training objective is

$$\mathcal{L}_{\text{train}} = \mathbb{E}_{s\sim d_{\mathcal{D}}}\left[D_{\text{KL}}(\pi^*(\cdot|s) \parallel \pi_\theta(\cdot|s))\right] \tag{10}$$

At inference, however, the student acts according to its own learned policy π_θ. Because the student is imperfect, its actions deviate from the dataset distribution. Let $d_{\pi_\theta}(s)$ denote the state visitation distribution induced by the student's own rollouts. The true generation quality is evaluated under this distinct distribution

$$\mathcal{L}_{\text{test}} = \mathbb{E}_{s\sim d_{\pi_\theta}}\left[\mathcal{L}_{\text{task}}(s, \pi_\theta)\right] \tag{11}$$

Exposure bias arises from the inequality $d_{\mathcal{D}}(s) \neq d_{\pi_\theta}(s)$. Because the student never encounters states in $d_{\pi_\theta}(s) \setminus d_{\mathcal{D}}(s)$ during training, it receives zero supervisory signal in these regions.

We quantify this mismatch using performance bounds from imitation learning. As Ross et al. (2011) showed via the DAgger analysis, if a student policy π_θ mimics a teacher policy π^* with per-step error bounded by ϵ under the training distribution, i.e., $\mathbb{E}_{s\sim d_{\pi^*}}[\mathbb{I}(\pi_\theta(s) \neq \pi^*(s))] \leq \epsilon$, then the expected total discrepancy over a trajectory of length T under the student's own distribution scales quadratically

$$\mathbb{E}_{s\sim d_{\pi_\theta}}\left[\sum_{t=1}^{T} \mathcal{L}(s_t)\right] \leq O(\epsilon T^2) \tag{12}$$

This $O(T^2)$ compounding bound is significant for modern LLMs, which routinely generate sequences spanning thousands of tokens. A single suboptimal token shifts the prefix slightly out-of-distribution, and the model, having never seen this perturbed prefix, is more likely to err again, leading to degraded or hallucinatory text. On-policy distillation resolves this by changing the training expectation from $\mathbb{E}_{s\sim d_{\mathcal{D}}}$ to $\mathbb{E}_{s\sim d_{\pi_\theta}}$. By generating responses online, the student confronts its own mistakes, receives teacher feedback on those specific out-of-distribution states, and learns robust recovery behaviors, reducing error accumulation from $O(\epsilon T^2)$ to $O(\epsilon T)$.

Remark: The Nuance of the DAgger Bound in LLMs. While the theoretical transition from $O(\epsilon T^2)$ to $O(\epsilon T)$ is appealing, applying the DAgger bound to LLMs requires nuance. The original DAgger theorem assumes an *interactive expert* providing optimal actions in any state. In white-box OPD, the "expert" provides a next-token distribution $p_T(y_t|\hat{y}_{<t})$ conditioned on the student's prefix $\hat{y}_{<t}$. If the student hallucinates a severely out-of-distribution prefix, the teacher's conditional distribution may itself become poorly calibrated. Forcing the student to match this noisy distribution violates the core assumption of interactive imitation learning, potentially destabilizing training rather than recovering the $O(\epsilon T)$ bound. This underscores why adaptive divergence methods (Section 4.1) are critical for selectively trusting the teacher.

This imitation learning perspective (train on student rollouts, supervise with teacher feedback) directly inspired the modern LLM distillation methods. GKD (Agarwal et al., 2024) and

DistiLLM (Ko et al., 2024) generalized this DAgger-style pattern to the large language model setting, replacing neural machine translation-scale architectures with billion-parameter models and introducing divergence-aware training objectives.

2.4 The f-Divergence Family and Geometric Intuition

The exposure bias analysis above establishes *why* training on the student's own rollouts is necessary, but leaves open the question of *how* to measure the teacher–student discrepancy once on-policy trajectories are available. Different distance measures impose qualitatively different geometric constraints on the student's learned distribution, and the choice among them is the single most consequential design decision in token-level OPD.

To parameterize distribution matching in OPD, we rely on the f-divergence framework. Given two probability distributions P and Q over a space $\mathcal{X}$, an f-divergence measures their discrepancy via a convex generator $f : (0, \infty) \to \mathbb{R}$ with $f(1) = 0$

$$D_f(P \parallel Q) = \int_{\mathcal{X}} Q(x) f\left(\frac{P(x)}{Q(x)}\right) dx = \mathbb{E}_{x\sim Q}\left[f\left(\frac{P(x)}{Q(x)}\right)\right] \tag{13}$$

The choice of $f(u)$ yields different properties for the distillation objective. In LLM distillation, we identify $P \equiv p_T$ (teacher) and $Q \equiv p_\theta$ (student), where the token-level notation matches our convention from Section 2.1.

1. Forward Kullback-Leibler Divergence. Selecting $f(u) = u \log u$ recovers Forward KL, $D_{\text{KL}}(P \parallel Q) = \mathbb{E}_{x\sim P}[\log(P(x)/Q(x))]$. The gradient forces the student Q to place mass everywhere the teacher P has mass. If $P(x) > 0$ but $Q(x) \approx 0$, the penalty diverges. Forward KL is thus strongly *mode-covering* (zero-avoiding). In LLM distillation, because a small student cannot memorize the teacher's vast distribution, covering all modes spreads probability into the void between valid modes, producing nonsensical, average phrasing.

2. Reverse Kullback-Leibler Divergence. Selecting $f(u) = -\log u$ recovers Reverse KL, $D_{\text{KL}}(Q \parallel P) = \mathbb{E}_{x\sim Q}[\log(Q(x)/P(x))]$. The expectation is over the student Q. If $Q(x) > 0$ where $P(x) \approx 0$, the penalty diverges. The student therefore assigns probability only to regions supported by the teacher, yielding *mode-seeking* (zero-forcing) behavior, collapsing onto one major mode while ignoring minor ones. For generative language, this is desirable because producing one coherent answer is preferable to an incoherent amalgamation of multiple correct answers.

3. Jensen-Shannon Divergence (JSD). JSD introduces symmetry by measuring divergence to a mixture $M = \frac{1}{2}(P + Q)$. Using $f(u) = \frac{1}{2}\left[u \log\left(\frac{2u}{u+1}\right) + \log\left(\frac{2}{u+1}\right)\right]$, JSD is bounded in $[0, \log 2]$, offering a stable gradient field that balances mode-seeking and mode-covering behavior, preventing extreme gradient explosions in out-of-distribution regions.

4. Total Variation Distance (TVD). Using $f(u) = \frac{1}{2}|u - 1|$, TVD measures the maximum absolute probability difference over any event, $\sup_A |P(A) - Q(A)|$. TVD provides a robust distance metric irrespective of extreme log-probability ratios, though its non-differentiability at $u = 1$ challenges direct optimization in neural spaces. The choice from this f-divergence family dictates how the student interpolates teacher knowledge during on-policy generation.

2.5 A Unified Mathematical View of Modern OPD

Recent OPD methods can be unified under a single objective by decoupling the sampling distribution (which dictates the trajectory) from the divergence metric (which dictates local token-level matching). The generalized OPD objective is

$$\mathcal{L}_{\text{OPD}}(\theta) = \mathbb{E}_{y\sim\pi_{\text{mix}}}\left[\sum_{t=1}^{|y|} \mathcal{D}_f\left(p_T(\cdot|x, y_{<t}),\ p_\theta(\cdot|x, y_{<t})\right)\right] \tag{14}$$

where π_{mix} is the behavioral policy driving state exploration and $\mathcal{D}_f(\cdot, \cdot)$ is a divergence from the f-divergence family. We use the two-argument notation $\mathcal{D}_f(P, Q)$ rather than

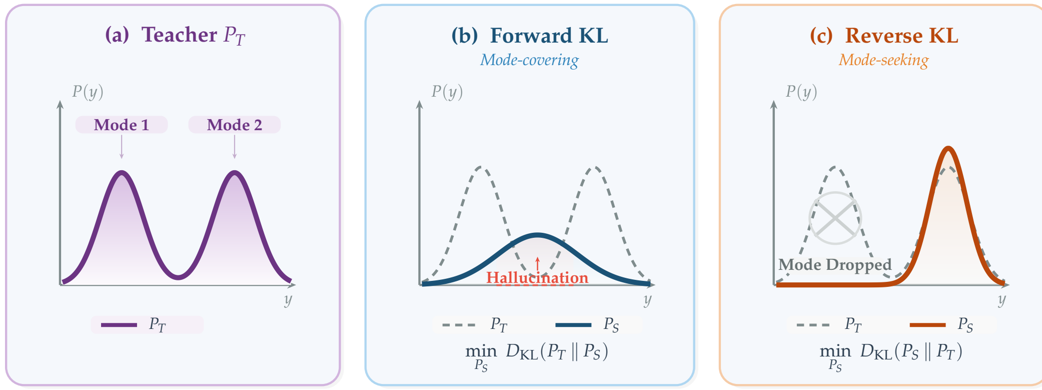

Figure 1: Forward KL vs. Reverse KL divergence for fitting a student distribution P_S to a bimodal teacher P_T. (a) The teacher distribution with two modes. (b) Forward KL is mode-covering (zero-avoiding): the student covers both modes but places mass in the inter-mode "hallucination zone." (c) Reverse KL is mode-seeking (zero-forcing): the student concentrates on a single peak, dropping the other mode entirely. Adaptive methods (ToDi, Entropy-Aware OPD) dynamically switch between the two based on teacher confidence.

$\mathcal{D}_f(P\|Q)$ because different methods place the teacher and student in different argument positions: Forward KL computes $D_{\mathrm{KL}}(p_T\|p_\theta)$ (teacher-first), while Reverse KL computes $D_{\mathrm{KL}}(p_\theta\|p_T)$ (student-first). The key modern methods (GKD (Agarwal et al., 2024), MiniLLM (Gu et al., 2024), and DistiLLM (Ko et al., 2024)) all map onto this equation as distinct parameterizations of a continuous theoretical space, differing in the choice of f, the argument ordering, and the target distribution.

Generalized Knowledge Distillation (GKD) (Agarwal et al., 2024)**.** GKD addresses trajectory mismatch by defining π_{mix} as an explicit interpolation between the dataset and the student's on-policy distribution. The output sequence is drawn with probability λ from p_θ and with probability $1-\lambda$ from $\mathcal{D}$. GKD is flexible regarding $\mathcal{D}_f$, empirically testing Forward KL, Reverse KL, and JSD. Setting $\lambda \to 1$ makes GKD purely on-policy.

MiniLLM (Gu et al., 2024)**.** MiniLLM identifies that Forward KL forces over-allocation to the teacher's long tail. To enforce mode-seeking behavior, it selects Reverse KL ($\mathcal{D}_f = D_{\mathrm{KL}}(p_\theta \parallel p_T)$). For sampling, MiniLLM employs a teacher-mixed strategy, where with probability $\alpha = 0.2$, tokens are drawn from the teacher to alleviate reward hacking, making $\pi_{\text{mix}} = (1-\alpha)p_\theta + \alpha p_T$. Because Reverse KL places the student parameters in both the expectation and the log-ratio, MiniLLM reformulates optimization via the Policy Gradient theorem (REINFORCE), treating the teacher's log-probability as a reward. Defining the cumulative reward-to-go $R_t = \sum_{t'=t}^{|y|} \log \frac{p_T(y_{t'}|y_{<t'})}{p_\theta(y_{t'}|y_{<t'})}$, the gradient becomes

$$\nabla_\theta \mathcal{L}_{\text{MiniLLM}} = -\mathbb{E}_{y\sim p_\theta}\left[\sum_{t=1}^{|y|} (R_t - 1)\, \nabla_\theta \log p_\theta(y_t|y_{<t})\right] \tag{15}$$

To mitigate the high variance of policy gradients, MiniLLM decomposes R_t into a single-step reward $r_t = \log \frac{p_T(y_t|y_{<t})}{p_\theta(y_t|y_{<t})}$ plus a future return, enabling per-token baselines that bridge RL optimization with knowledge distillation.

DistiLLM (Ko et al., 2024)**.** While MiniLLM theoretically achieves mode-seeking behavior, reliance on high-variance RL causes empirical instability. DistiLLM resolves this with a pair of *skewed* KL divergences. The primary objective uses Skewed Forward KL: it defines a skewed mixture $\tilde{p} = \alpha p_T + (1-\alpha)p_\theta$ and minimizes $D_{\mathrm{KL}}(p_T \parallel \tilde{p})$, which avoids the zero-division instability of standard Forward KL when $p_\theta(y) \approx 0$, because the mixture $\tilde{p}$ is always at least $\alpha p_T(y) > 0$. Since the expectation is over the fixed teacher p_T, the loss reduces to a tractable cross-entropy without policy gradients. DistiLLM additionally employs a complementary Skewed Reverse KL, $D_{\mathrm{KL}}(p_\theta \parallel (1-\alpha)p_T + \alpha p_\theta)$, on student-generated outputs, and the combination of both losses is shown to be synergistic. Under our unified view, DistiLLM maintains $\pi_{\text{mix}} = p_\theta$ and replaces the standard KL target from p_T to skewed mixtures, achieving tractability and stability.

This unified perspective demonstrates that modern OPD is not a series of disparate heuristics, but a systematic, progressive exploration of a design space governed by three interacting choices: trajectory sampling (π_{mix}), divergence generator (f), and the argument ordering and target distribution within the chosen divergence.

3 Taxonomy and Organization

With the mathematical foundations in place, we now turn to the question of how to organize the rapidly growing OPD literature. A useful taxonomy must balance two competing demands: it should impose enough structure to reveal meaningful clusters and underexplored regions of the design space, yet remain flexible enough to accommodate the hybrid methods that increasingly blur traditional category boundaries. This section presents the organizational framework for the remainder of this survey. We first motivate a binary top-level split that partitions OPD methods by the role of the teacher signal (Section 3.1), then describe a five-dimensional annotation scheme that captures orthogonal design choices within each category (Section 3.2), and finally provide deeper exposition of three dimensions that most directly determine algorithmic behavior (Section 3.3).

3.1 Organization Rationale

We organize the OPD literature along a single top-level binary distinction: *teacher-guided* methods (Section 4), which rely on an external teacher model (whether through full logit access, black-box API queries, or reward-guided signals) and *self-distillation* methods (Section 5), where the model bootstraps its own capabilities without a separate, superior teacher. This split is stable and unambiguous: every method either requires an external teacher or does not. Within each category, subsections are organized by *technical challenge* rather than by chronological release or minor architectural variation, reflecting the conceptual threads that drive the field's evolution.

3.2 Multi-Dimensional Characterization

Rather than assigning each method to a single mutually exclusive category, we annotate methods along five soft dimensions that capture orthogonal design choices: (1) **teacher_source** (white-box logit access, black-box API, or self/teacher-free), (2) **signal_type** (logit-based distributional matching, outcome-based scalar or preference signals, or self-play), (3) **granularity** (token-level, sequence-level, or hybrid/adaptive), (4) **uses_rl** (whether the method incorporates reinforcement learning objectives), and (5) **divergence** (the specific f-divergence or loss function employed). A single method may span multiple values on each dimension. For instance, SDPO (Hübotter et al., 2026) is self-distillation by teacher_source, uses textual feedback as signal_type, operates at hybrid granularity, and incorporates RL-style credit assignment. These dimensions are not mutually exclusive categories but complementary lenses for understanding the design space. Table 1 illustrates the multi-dimensional annotation for representative methods spanning all major sections of this survey.

3.3 Design Dimensions Within the Binary Split

The binary split and five-dimensional annotation introduced above define *where* a method sits in the design space. Three of these dimensions deserve deeper exposition because they most directly determine algorithmic behavior and practical deployment constraints. Understanding how these dimensions interact within each branch of the binary taxonomy is essential for selecting the right method for a given resource budget and task profile.

Table 1: Multi-dimensional annotation of representative on-policy distillation methods. Teacher Source: External (white-box or API access to a separate teacher) vs. Self (no separate teacher). Signal Type: the primary supervisory signal. Granularity: token-level, sequence-level, or hybrid. RL: whether reinforcement learning objectives are incorporated. Divergence/Loss: the specific optimization objective.

Method	**§**	**Teacher Source**	**Signal Type**	**Granularity**	**RL**	**Divergence / Loss**
Teacher-Guided On-Policy Distillation (Section 4)						
GKD (Agarwal et al., 2024)	4.1	External	Logit	Token	–	JSD / FKL / RKL
MiniLLM (Gu et al., 2024)	4.1	External	Logit	Sequence	✓	Reverse KL
DistiLLM (Ko et al., 2024)	4.1	External	Logit	Token	–	Skewed KL
ToDi (Jung et al., 2025)	4.1	External	Logit	Token	–	Per-token adaptive FKL+RKL
Entropy-Aware (Jin et al., 2026)	4.1	External	Logit	Token	–	Entropy-gated FKL+RKL
G-OPD (Yang et al., 2026c)	4.1	External	Logit	Token	✓	KL-constrained RL
PACED (Xu et al., 2026a)	4.2	External	Logit	Hybrid	–	Beta-kernel weighted KL
Fast OPD (Zhang et al., 2026a)	4.2	External	Logit	Hybrid	–	Reverse KL on prefix
DSKD (Zhang et al., 2025b)	4.3	External	Logit	Token	–	Dual-space KL
GAD (Ye et al., 2025)	4.4	External	Adversarial	Sequence	✓	Adversarial minimax
Lion (Jiang et al., 2023)	4.4	External	Verbal	Sequence	–	Adversarial curriculum
RLKD (Xu et al., 2025b)	4.5	External	Reward	Sequence	✓	KL-regularized RL + GSRM
AlignDistil (Zhang et al., 2025a)	4.5	External	Preference	Hybrid	–	DPO + token-level KD
RLAD (Zhang et al., 2026e)	4.5	External	Logit+Reward	Token	✓	Trust-region ratio distillation
Self-Distillation (Section 5)						
SPIN (Chen et al., 2024)	5.2	Self	Self-play	Sequence	–	Self-play DPO
OPSD (Zhao et al., 2026b)	5.1	Self (PI)	Logit	Token	–	FKL / JSD
GATES (Stein et al., 2026)	5.1	Self (PI)	Logit	Token	–	Consensus-gated KL
SD-ZERO (He et al., 2026)	5.3	Self	Logit	Token	–	Revision-guided Reverse KL
CRISP (Sang et al., 2026)	5.1	Self (PI)	Logit	Token	–	Per-token Reverse KL
SDPO (Hübotter et al., 2026)	5.3	Self	Logit+Verbal	Token	✓	Self-distill + credit assign.
SRPO (Li et al., 2026b)	5.3	Self	Reward	Sequence	✓	GRPO + self-refinement
π-Play (Zhang et al., 2026d)	5.2	Self (QCP-PI)	Logit	Token	✓	Multi-agent self-distill
Applications & Systems (Section 7)						
Qwen3 (Yang et al., 2025a)	7.1	External (multi)	Logit	Token	✓	Ensemble distillation
ORBIT (Liang et al., 2026)	7.1	External (multi)	Logit	Token	✓	Multi-teacher mode fusion
VOLD (Bousselham et al., 2025)	7.3	External (text)	Logit+Reward	Token	✓	Cross-modal KL + GRPO

Signal type. Within teacher-guided OPD, the choice of supervisory signal largely determines training stability and information throughput. Logit-based feedback provides the densest gradient signal, evaluating the student's generated sequence token-by-token against the teacher's continuous probability distribution so the loss vector has size $|V|$ at every step t (Agarwal et al., 2024; Gu et al., 2024; Ko et al., 2024). Outcome-based feedback replaces exact token matching with scalar reward signals evaluating the correctness of a generated trajectory, granting the student freedom to find novel paths to correctness without mimicking the teacher's specific phrasing (Xu et al., 2025b; Zhang et al., 2025a). In self-distillation, the analogous choice arises between self-play feedback that constructs supervision from the student's own evolving policy (Chen et al., 2024) and privileged-information signals derived from an augmented version of the same model (Zhao et al., 2026b).

Teacher source. This dimension refines the binary split by further differentiating the nature of teacher access within each category. Among teacher-guided methods, white-box distillation allows exact analytical divergence matching using the full logit tensor, exploiting dark knowledge in minuscule probabilities on incorrect tokens (Agarwal et al., 2024; Ko et al., 2024). Black-box distillation must creatively approximate the teacher distribution when only API-level access is available (Ye et al., 2025; Jiang et al., 2023). Within self-distillation, the "teacher" may be a stabilized historical checkpoint, a privileged-information-conditioned variant, or an ensembled version of the student itself (Chen et al., 2024; Zhao et al., 2026b; Stein et al., 2026).

Granularity. This dimension cuts across both branches of the taxonomy. Token-level methods provide immediate, localized feedback at every timestep but suffer from myopia. Sequence-level methods apply the signal only at generation termination via outcome reward models, granting freedom but creating acute credit-assignment difficulty. Hybrid and adaptive approaches bridge token myopia and sequence sparsity by combining dense token-level supervision with step-level or trajectory-level reward signals (Lightman et al., 2024; Yang et al., 2025b; Xu et al., 2025b; Cai & Yuan, 2026). Process Reward Models

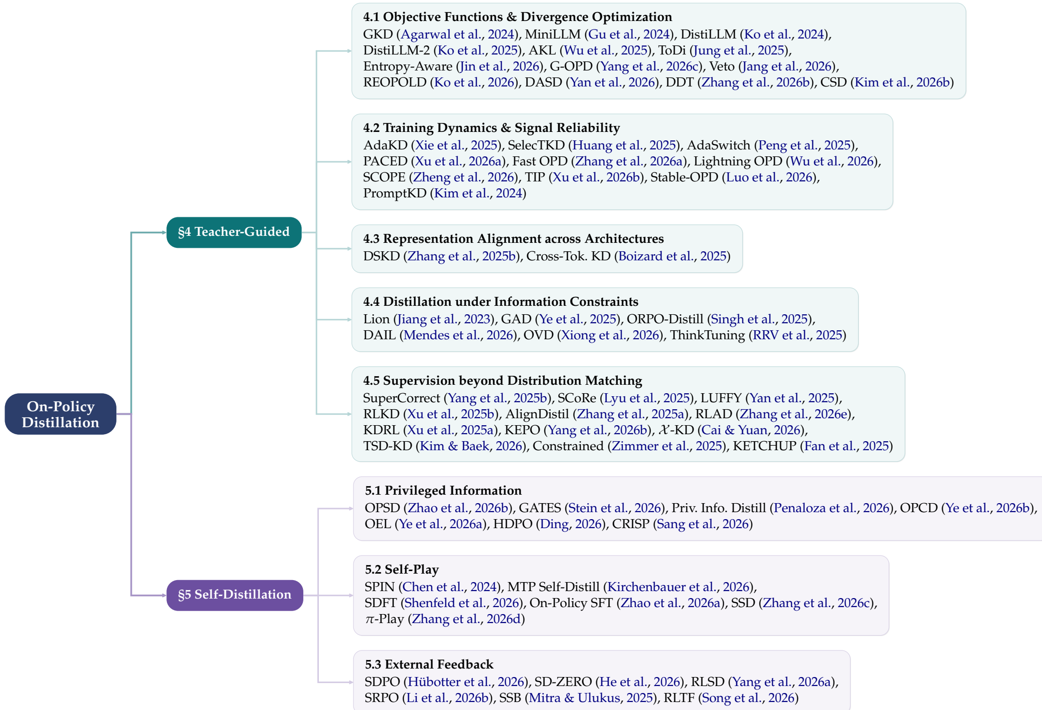

Figure 2: Taxonomy of on-policy distillation methods for large language models. Methods are organized by the primary role of the teacher signal: **Teacher-Guided OPD** (§4), where an external or frozen teacher provides supervision, and **Self-Distillation** (§5), where the student serves as its own teacher. Bracketed numbers correspond to the bibliography.

(PRMs) (Lightman et al., 2024) provide per-step verification of intermediate reasoning, enabling fine-grained credit assignment that neither pure token-level nor pure sequence-level approaches achieve. SuperCorrect (Yang et al., 2025b) leverages hierarchical thought templates extracted from the teacher to guide step-level reasoning distillation, combined with cross-model DPO for self-correction. These hybrid granularity approaches balance the stability of dense token feedback with the structural awareness of step-level evaluation, enabling more effective distillation of complex reasoning chains.

The resulting organizational structure is visualized in Figure 2. The primary binary split forms the backbone of Sections 4 and 5, while the multi-dimensional annotations in Table 1 enable cross-cutting comparisons that surface underexplored regions of the design space.

Tables 2 and 3 (placed after Section 4) provide comprehensive method comparisons and experimental configurations across all surveyed approaches.

4 Teacher-Guided On-Policy Distillation

Teacher-guided on-policy distillation is the most extensively studied branch of the OPD landscape, combining the rich supervisory signal of an external teacher with the distributional benefits of student-generated trajectories. The defining feature of this branch is the presence of a distinct, typically larger teacher model whose knowledge the student seeks to absorb. Traditional off-policy KD forces the student to mimic the teacher under the teacher's marginal $p_T(x, y)$, whereas OPD returns sampling authority to the student's own evolving policy $p_\theta(x, y)$. This shift eliminates exposure bias but introduces the challenge of efficiently and stably aligning the teacher's signals over trajectories that the student, not the teacher, has generated. In contrast, the self-play and self-distillation methods surveyed in Section 5 dispense with an external teacher altogether, instead deriving supervisory signal from the student's own outputs under privileged or augmented conditions.

The remainder of this section is organized by five principal technical challenges. We begin with the most fundamental design axis, objective function design and the evolution from fixed to adaptive divergences (Section 4.1). Once the divergence is chosen, practical training introduces its own difficulties, so Section 4.2 examines how token weighting, compute efficiency, curriculum design, and capacity gap bridging shape real-world outcomes. When teacher and student belong to different model families, vocabulary and representation mismatches must be resolved before any divergence can be computed, a problem addressed in Section 4.3. The challenge intensifies further when even the teacher's logits are inaccessible, and Section 4.4 covers distillation under such information constraints. Finally, Section 4.5 surveys methods that move beyond pure distribution matching by integrating reinforcement learning, chain-of-thought transfer, and structured reward signals.

4.1 Objective Functions and Divergence Optimization

Among all design choices in on-policy distillation, the divergence function occupies a uniquely foundational role: it determines whether the student's learned distribution will be mode-covering (spreading probability across all teacher modes), mode-seeking (collapsing onto the dominant mode), or somewhere between. As established in the unified framework of Section 2.5, all token-level methods instantiate a common objective parameterized by a sampling policy π_{mix} and an f-divergence $\mathcal{D}_f$. The divergence is the first knob a practitioner turns, and its choice cascades into every downstream behavior, from gradient stability to mode diversity. The field has thus progressed from fixed, globally applied divergences toward increasingly adaptive, context-sensitive formulations that vary the matching criterion within a single sequence. This subsection traces that evolution, beginning with uniform divergences (GKD, DistiLLM), advancing through per-token adaptive methods (ToDi, Entropy-Aware OPD), and concluding with sequence-level alternatives that recast distillation as reinforcement learning.

4.1.1 From Fixed to Adaptive Divergences

Baseline: uniform divergence with on-policy sampling. GKD (Agarwal et al., 2024) established the canonical on-policy distillation framework by introducing a mixture sampling policy π_{mix} that interpolates between the dataset and the student's own generations, controlled by a student data fraction $\lambda \in [0, 1]$:

$$\mathcal{L}_{GKD} = \mathbb{E}_{x\sim\mathcal{D},\, y\sim\pi_{\text{mix}}(\cdot|x)} \left[\sum_{t=1}^{|y|} D\big(p_T(\cdot|x, y_{<t}) \parallel p_\theta(\cdot|x, y_{<t})\big) \right] \tag{16}$$

where D can be Forward KL, Reverse KL, or generalized JSD with parameter β. GKD evaluates all three families and recommends Reverse KL or JSD for RL integration due to their bounded gradients, while using Forward KL for supervised settings. Setting $\lambda = 1$ yields purely on-policy training, while $\lambda = 0$ recovers standard off-policy KD. While GKD substantially mitigates compounding errors compared to off-policy baselines (Agarwal et al., 2024), a single globally fixed divergence cannot adapt to the varying difficulty levels across tokens in a sequence.

Stabilizing Reverse KL via interpolation. The uniform-divergence limitation of GKD becomes acute when the teacher and student probability masses are far apart, causing gradient instabilities under standard KL. DistiLLM (Ko et al., 2024) addressed this failure mode by constructing a skewed mixture target $\tilde{p} = \alpha p_T + (1-\alpha) p_\theta$ that guarantees bounded density ratios. The Skewed Forward KL (SKL) objective is:

$$\mathcal{L}_{\text{SKL}} = \mathbb{E}_{y\sim p_\theta} \left[\sum_t D_{\text{KL}}\big(p_T(\cdot|y_{<t}) \parallel \alpha p_T(\cdot|y_{<t}) + (1-\alpha) p_\theta(\cdot|y_{<t})\big) \right] \tag{17}$$

DistiLLM combines student-generated outputs (SGOs) with an adaptive off-policy replay buffer, using p_θ for generation and replaying past SGOs for sample efficiency. Since the inner KL expectation is taken over the fixed teacher p_T, the per-token loss reduces to a tractable cross-entropy $-\sum p_T \log \tilde{p}$ whose gradient with respect to θ flows analytically through the mixture $\tilde{p}$, avoiding the high-variance REINFORCE estimation that

MiniLLM requires. DistiLLM additionally uses a complementary Skewed Reverse KL (SRKL), $D_{\mathrm{KL}}(p_\theta \| (1-\alpha)p_T + \alpha p_\theta)$, on student-generated outputs, and the synergy of both losses further improves convergence. Building on this foundation, DistiLLM-2 (Ko et al., 2025) observes that teacher-generated and student-generated outputs benefit from *different* loss functions: Skew Forward KL (SKL) is applied to teacher outputs to increase the student's likelihood of high-quality responses, while Skew Reverse KL (SRKL) is applied to student outputs to decrease the likelihood of lower-quality generations. This contrastive asymmetry, combined with curriculum-based adaptive coefficients and speculative-decoding-based dataset curation, achieves state-of-the-art results among token-level distillation methods across instruction following, code generation, and mathematical reasoning.

Per-token adaptive divergence. GKD and DistiLLM both improve upon naive off-policy KD, yet they share a common limitation: a single divergence is applied uniformly across all tokens, ignoring the heterogeneous difficulty landscape within a sequence. Although not itself an on-policy method, ToDi (Jung et al., 2025) introduces a per-token adaptive divergence mechanism that directly inspired subsequent on-policy work. ToDi argues that different positions in a sequence demand different matching strictness: critical reasoning pivots require strict Reverse KL, while syntactic fillers tolerate relaxed Forward KL. The method adaptively combines Forward and Reverse KL at each position using a sigmoid-based weighting derived from the teacher–student log-probability ratio:

$$\mathcal{L}_{ToDi} = \mathbb{E}_{y\sim\mathcal{D}}\left[\sum_t \omega_t D_{f^{(t)}}\big(p_T(\cdot|y_{<t}) \parallel p_\theta(\cdot|y_{<t})\big)\right] \tag{18}$$

where $y \sim \mathcal{D}$ reflects ToDi's original off-policy formulation (training on dataset sequences). The adaptive weighting ω_t is orthogonal to the sampling strategy and transfers directly to on-policy settings. When the ratio indicates that the student underestimates the teacher's high-confidence predictions, Reverse KL dominates to suppress errors. When the teacher's distribution is more uncertain, Forward KL contributes more to preserve diversity. The resulting per-token adaptability represents a qualitative advance over globally fixed divergences, though it introduces architectural complexity through the heuristic computation of ω_t.

Balancing head and tail of the distribution. While ToDi adapts the divergence based on positional signals, AKL (Wu et al., 2025) approaches the same problem from a distributional perspective. Also developed in an off-policy context, AKL challenges the conventional wisdom that Reverse KL is strictly superior for LLM distillation. Through both theoretical analysis and experiments, Wu et al. (2025) demonstrate that the mode-seeking and mode-covering characterizations of Reverse and Forward KL do not strictly hold for discrete distributions in LLMs, and that both divergences converge to the same objective given sufficient training epochs. Under practical epoch budgets, however, Forward KL focuses learning on the head of the teacher distribution while Reverse KL focuses on the tail. AKL adaptively allocates weights to combine both divergences based on the current teacher and student distributions, achieving superior alignment under limited training budgets. This head-tail analysis complements ToDi's per-token adaptation by providing a distributional rather than positional perspective on divergence selection. We include both ToDi and AKL in this section because their adaptive weighting mechanisms are directly applicable to on-policy sampling and have inspired subsequent on-policy methods such as Entropy-Aware OPD (Jin et al., 2026).

Entropy-gated divergence switching. The adaptive mechanisms of ToDi and AKL rely on heuristic weight computations, and neither was originally formulated for the on-policy setting. Entropy-Aware OPD (Jin et al., 2026) bridges this gap by identifying a complementary failure mode specific to on-policy training: when the teacher distribution has high entropy (genuine uncertainty about the correct continuation), Reverse KL's mode-seeking property forces the student to arbitrarily select one of many equally plausible modes, discarding valuable distributional information and yielding unstable learning signals. The solution is an entropy-gated mixture:

$$\mathcal{L}_{Ent\text{-}OPD} = \mathbb{E}_{y\sim p_\theta}\Big[\sum_t (1-\alpha_t) D_{\mathrm{KL}}\big(p_\theta(\cdot|y_{<t}) \parallel p_T(\cdot|y_{<t})\big) + \alpha_t D_{\mathrm{KL}}\big(p_T(\cdot|y_{<t}) \parallel p_\theta(\cdot|y_{<t})\big)\Big] \tag{19}$$

where α_t increases with teacher entropy $\mathcal{H}(p_T(\cdot|y_{<t}))$. In low-entropy regions (confident predictions), Reverse KL dominates for precise imitation, while in high-entropy regions, Forward KL captures the full range of plausible outputs. On Qwen3-4B-Base, this achieves a +5.05 Pass@8 improvement over standard Reverse KL on mathematical reasoning, making it one of the most robust token-level strategies against distribution shifts, though it requires computing teacher entropy at every step.

OPD as KL-constrained reinforcement learning. The preceding methods all treat divergence minimization as the end goal. G-OPD (Yang et al., 2026c) reframes the problem entirely by proving that standard OPD is a special case of dense KL-constrained RL: minimizing the on-policy KL divergence between teacher and student is equivalent to maximizing a token-level reward (derived from the teacher's log-probabilities) subject to a KL penalty against a reference policy. By decoupling the reference model from the teacher and introducing a reward scaling factor α, G-OPD generalizes the framework. When $\alpha = 1$, it recovers standard OPD, and when $\alpha > 1$ (ExOPD), the student is incentivized to *exceed* the teacher's distribution, breaking the imitation ceiling that constrains standard distillation. Empirically, ExOPD enables student models to surpass teacher performance on mathematical reasoning benchmarks. **Why different divergences for different methods?** The divergence choices surveyed above are not arbitrary, as each reflects a deliberate response to specific failure modes. GKD (Agarwal et al., 2024) defaults to JSD for bounded, symmetric gradients. DistiLLM (Ko et al., 2024) adopts skewed KL divergences (both Forward and Reverse variants) because standard KL produces gradient instabilities when there are large probability mismatches between teacher and student. ToDi (Jung et al., 2025) and Entropy-Aware OPD (Jin et al., 2026) argue that the optimal divergence varies *within* a single sequence and adapt accordingly. Empirically, these design choices manifest in task-specific advantages: on mathematical reasoning benchmarks (GSM8K, MATH), Reverse KL variants consistently outperform Forward KL because mode-seeking behavior prevents the student from averaging over distinct solution strategies. On open-ended generation tasks (MT-Bench, AlpacaEval), Forward KL and JSD preserve the diversity needed for creative and conversational responses. Practitioners should match their divergence choice to the downstream task's tolerance for mode-dropping versus hallucination (see Figure 1).

A deeper geometric view. The Forward/Reverse KL dichotomy, while instructive, obscures a subtler geometric reality (Wu et al., 2025; Zhong et al., 2024). Under Forward KL, the student's distribution is pulled toward the *centroid* of the teacher's probability mass, which for multimodal distributions lies in a low-density region between modes, explaining the hallucination pathology. Under Reverse KL, the student collapses onto a single high-density mode, yielding confident but narrow outputs. JSD traces a path that partially covers multiple modes without fully committing to any, explaining its empirically observed "jack-of-all-trades" behavior (Agarwal et al., 2024).

This geometric perspective also illuminates why adaptive divergences (ToDi, AKL, Entropy-Aware) outperform fixed ones: different token positions exhibit different distributional geometry, and no single divergence is optimal everywhere. We develop this insight in detail in Section 4.1.

A detailed, multidimensional comparison of white-box methods is presented in Table 2. Table 3 provides experimental configurations across all surveyed methods.

4.1.2 *Sequence-Level Distillation as Reinforcement Learning*

While token-level methods provide dense supervision, they inherently suffer from greedy optimization bias: each token is matched independently, ignoring the joint probability of the full sequence. Sequence-level OPD addresses this by reframing distillation as trajectory-level reinforcement learning, treating the complete output as a single decision and optimizing a global reward.

MiniLLM and the REINFORCE derivation. MiniLLM (Gu et al., 2024) serves as the foundational sequence-level method. As introduced in Section 2.5, MiniLLM selects Reverse KL to enforce mode-seeking behavior at the sequence level. The central challenge is computing the gradient of an expectation over p_θ when p_θ itself is parameterized by θ. Using the log-derivative trick (REINFORCE), the derivation proceeds as follows. Starting from $D_{\mathrm{KL}}(p_\theta \parallel p_T) = \mathbb{E}_{y\sim p_\theta}[\log p_\theta(y|x) - \log p_T(y|x)]$ and applying the product rule:

$$\begin{aligned}\nabla_\theta D_{\mathrm{KL}}(p_\theta \| p_T) &= \nabla_\theta \sum_y p_\theta(y|x) \left(\log p_\theta(y|x) - \log p_T(y|x)\right) \\ &= \sum_y p_\theta(y|x) \nabla_\theta \log p_\theta(y|x) \log \frac{p_\theta(y|x)}{p_T(y|x)} + \sum_y p_\theta(y|x) \nabla_\theta \log p_\theta(y|x) \qquad (20)\end{aligned}$$

Since $\sum_y p_\theta(y|x) = 1$ and its gradient vanishes, the final gradient simplifies to:

$$\nabla_\theta \mathcal{L}_{MiniLLM} = \mathbb{E}_{y\sim p_\theta}\left[\Big(\log p_\theta(y|x) - \log p_T(y|x)\Big) \nabla_\theta \log p_\theta(y|x)\right] \qquad (21)$$

This shows that minimizing sequence-level Reverse KL is mathematically equivalent to policy gradient RL with reward $r(x, y) = \log p_T(y|x) - \log p_\theta(y|x)$. MiniLLM decomposes this into per-token rewards and future returns, enabling variance-reducing baselines. However, REINFORCE estimation in massive combinatorial output spaces demands substantially more training iterations and sophisticated baseline subtraction to converge.

Constrained optimization for stability. To combat the high variance of REINFORCE, Zimmer et al. (2025) recast distillation as a constrained Markov Decision Process (CMDP). Instead of treating the KL penalty as a soft Lagrangian term with manually tuned λ, they maximize task reward subject to a hard constraint $D_{\mathrm{KL}}(p_\theta \parallel p_T) \leq \epsilon$. By adapting constrained state-augmented RL, the method maintains theoretical guarantees of constraint satisfaction without requiring the teacher at deployment time. KETCHUP (Fan et al., 2025) addresses the variance problem from a different angle by inducing a K-step return via the Bellman Optimality Equation, replacing the high-variance single-step REINFORCE estimator with a multi-step formulation that provably reduces gradient variance, yielding more stable RL-based distillation especially for larger student models.

Token-level vs. sequence-level: the fundamental tradeoff. The contrast between these two granularities exposes a core tension in OPD design. Token-level methods optimize an upper bound of the sequence-level divergence, and their enforced local alignment yields low variance and high stability, but greedy optimization can perfect local syntax while missing global coherence. Sequence-level methods directly model trajectory reward and thus capture holistic reasoning structure, but the high variance of Monte Carlo sampling demands substantially more compute. This tradeoff motivates the hybrid approaches presented in Section 4.2, which blend both granularities to balance stability and expressiveness.

4.1.3 Alternative Objective Formulations

The methods above all operate within the KL divergence family, varying only the direction, weighting, or adaptivity of the divergence. A natural question arises: are probability-based divergences the right objective at all?

Concrete Score Distillation (CSD) (Kim et al., 2026b) challenges the softmax-based probability matching paradigm entirely. By operating directly on logits rather than probabilities, CSD avoids the information loss introduced by softmax normalization and captures richer structural relationships in the teacher's output distribution. The CSD objective provides complementary gains when integrated with on-policy techniques such as GKD and DistiLLM.

Veto (Jang et al., 2026) pursues a related goal from an on-policy perspective. Rather than mixing data samples, it constructs an intermediate target distribution through a geometric bridge in logit space. A tunable parameter β serves dual roles: as an *Adaptive Gradient Veto* that suppresses pathological gradients on low-confidence tokens (addressing Forward KL instability), and as a *Decisiveness Knob* that balances reward-driven performance with output diversity (addressing Reverse KL mode collapse). On reasoning and generation tasks, Veto outperforms both supervised fine-tuning and existing on-policy baselines including GKD and DistiLLM.

While $\mathcal{X}$-KD reconstructs the teacher's learning environment (discussed in detail in Section 4.5), REOPOLD (Ko et al., 2026) bridges OPD and policy optimization more directly by proving that the teacher–student log-likelihood ratio acts as a token-level reward. Building on this insight, it relaxes strict imitation constraints through mixture-based reward clipping (preventing over-trust of extreme teacher signals), entropy-based dynamic sampling (selectively leveraging teacher feedback at high-uncertainty tokens), and a unified exploration-to-refinement strategy. REOPOLD achieves 6.7–12$\times$ greater sample efficiency than recent RL approaches, enabling a 7B student to match a 32B teacher in visual reasoning with $\sim$3.32$\times$ inference speedup.

The remaining two methods in this category question not the divergence itself but the SFT-based distillation paradigm that sits around it. DASD (Yan et al., 2026) critically reexamines the dominant practice of SFT on teacher-generated reasoning traces, identifying three limitations: inadequate representation of the teacher's sequence-level distribution from single best-of-n samples, misalignment between teacher output distribution and student capacity, and exposure bias from teacher-forced training. Its enhanced distribution-aligned pipeline achieves state-of-the-art performance at comparable scale using only 448K training samples.

DDT (Zhang et al., 2026b) complements DASD's empirical findings with a theoretical framework explaining why RL generalizes better than SFT: the key factor is RL's on-policy data generation, which keeps training aligned with the model's evolving distribution. DDT proposes In-Distribution Finetuning (re-weighting losses to down-weight out-of-distribution samples) and Hinted Decoding (re-aligning the training corpus to the model's current distribution), achieving generalization surpassing Direct Preference Optimization (DPO) and SimPO while maintaining supervised fine-tuning (SFT) efficiency.

The common thread across these alternative formulations is a move from matching the teacher's probability distribution to matching its *intent*: CSD and Veto preserve structural information that softmax discards, REOPOLD reinterprets the teacher signal as a reward rather than a target, and DASD/DDT question whether the distribution should be matched at all versus used as a guide for in-distribution learning. This progression from distribution matching to intent matching mirrors the broader evolution of the OPD field.

4.1.4 *Divergence Theory*

The preceding subsubsections have presented divergence choices as engineering decisions motivated by empirical performance. We now synthesize the theoretical underpinnings that justify these choices and reveal deeper structural relationships among them.

Forward KL vs. Reverse KL: beyond the standard dichotomy. The conventional view holds that Forward KL is zero-avoiding (mode-covering), meaning the student spreads mass to cover all teacher modes at the risk of placing probability in inter-mode regions, while Reverse KL is zero-forcing (mode-seeking), concentrating on a single mode at the cost of diversity. However, Wu et al. (2025) challenge this dichotomy for discrete distributions in LLMs, demonstrating that the mode-seeking and mode-covering characterizations do not strictly hold, and that both divergences converge to the same objective given sufficient training epochs. Under practical epoch budgets, the difference is better understood through the lens of *head-tail focus*: Forward KL concentrates learning on the head of the teacher distribution, while Reverse KL emphasizes the tail. Zhong et al. (2024) provide complementary evidence that the choice between Forward and Reverse KL is task-dependent rather than universally prescribable, with adaptive combinations consistently outperforming either fixed choice.

The unified f-divergence perspective. Wen et al. (2023) showed that any f-divergence can be expressed as $D_f(p_T \parallel p_\theta) = \mathbb{E}_{y\sim p_\theta}[f(p_T(y)/p_\theta(y))]$ where f is convex with $f(1) = 0$. By continuously modulating the convexity parameter, researchers can transition smoothly between mode-seeking and mode-covering behavior, tailoring distillation to downstream requirements.

Why adaptive divergences win. The success of ToDi (Jung et al., 2025), AKL (Wu et al., 2025), and Entropy-Aware OPD (Jin et al., 2026) has a geometric explanation. At reasoning-critical tokens, the teacher's distribution is typically unimodal (one correct logical step), and Reverse KL aligns well. At generation-flexible tokens (choosing between "therefore" and "thus"), the distribution is near-uniform over synonyms, and Forward KL preserves this diversity. Adaptive methods implicitly detect this modality structure and switch accordingly. The practical implication is clear: no single divergence is optimal across all token positions, and the field has moved decisively from "which divergence?" to "which divergence *where*?"

4.2 Training Dynamics and Signal Reliability

The divergence optimization methods in the preceding subsection determine *what* the student should learn at each token position. Yet even with the ideal divergence, practical on-policy training introduces a distinct set of challenges: *which* tokens and trajectories carry the most learning signal, and how should the student generate them efficiently within a finite compute budget? In on-policy training, every student rollout incurs a non-trivial generation cost, making these questions first-order determinants of both wall-clock efficiency and final model quality rather than secondary implementation details. This subsection covers four complementary strategies for managing these practical bottlenecks, spanning token weighting and selection, compute efficiency, curriculum design, and capacity gap bridging.

4.2.1 Token Weighting and Selection

Standard OPD objectives treat all positions uniformly, yet the information content per token varies by orders of magnitude within a single sequence. A critical reasoning pivot (choosing between "multiply" and "add" in a math derivation) carries far more distillation signal than a syntactic filler ("therefore" vs. "thus"). This heterogeneity creates two inefficiencies: uninformative tokens dilute the gradient signal, and all rollouts receive equal treatment regardless of whether they expose genuine student weaknesses.

The first generation of solutions operates at the *token level*. AdaKD (Xie et al., 2025) upweights tokens where the student–teacher gap is large and the student's confidence is low, while SelecTKD (Huang et al., 2025) filters out tokens that contribute little to knowledge transfer. Both share a key insight: the optimal distillation weight at each position is a function of the student's *current* competence, not a static property of the data. This perspective reframes token weighting as an implicit curriculum that adapts within each training step.

TIP (Xu et al., 2026b) provides a principled theoretical foundation for this observation by organizing token importance along two axes already computed during standard OPD: student entropy h_t and teacher–student divergence δ_t. Crossing these axes yields four quadrants with qualitatively distinct learning roles: high-entropy tokens where the student is uncertain (Q1/Q2) carry the dominant corrective signal, while low-entropy, high-divergence tokens (Q3) represent *overconfident errors*, positions where the student is certain but wrong, that any entropy-only weighting scheme is provably blind to. The key theoretical result (Proposition 2 in the original paper) shows that this blindness is structural, not a matter of threshold tuning: because Q3 tokens have near-zero entropy by definition, no non-decreasing function of entropy alone (with the natural boundary condition $f(0) = 0$) can distinguish them from the genuinely solved Q4 tokens. TIP's parameter-free Soft-OR score, $s_t = \hat{h}_t + \hat{\delta}_t - \hat{h}_t \cdot \hat{\delta}_t$, recovers Q3 coverage while preserving Q1/Q2 sensitivity, requiring no hyperparameters and no computation beyond what standard OPD already performs. Across three model families (Qwen3, Llama, Qwen2.5) with capacity gaps ranging from 2× to 9×, Soft-OR at 50% token retention consistently outperforms both full-token OPD and entropy-only selection on mathematical reasoning, while reducing peak memory by up to 47%. TIP thus transforms the earlier token-weighting heuristics of AdaKD and SelecTKD into a theoretically grounded framework, explaining *why* those methods work (they predominantly capture Q1/Q2 tokens) and identifying *where* they fail (Q3 tokens, which are rare but carry disproportionate corrective signal).

SCOPE (Zheng et al., 2026) elevates this analysis from tokens to *rollouts*, revealing a subtler problem: not all student-generated trajectories carry equally reliable supervisory information. When the student produces a flawed prefix, the teacher may struggle to provide useful guidance because the teacher itself has never been conditioned on such out-of-distribution context, a phenomenon SCOPE terms the "flawed prefix trap." The solution is a dual-path architecture that routes rollouts by correctness: incorrect trajectories receive teacher-perplexity-weighted KL distillation (down-weighting unreliable guidance), while correct trajectories receive student-perplexity-weighted MLE (concentrating reinforcement at the capability boundary). This design directly addresses diversity collapse, the counterintuitive phenomenon where Pass@1 improves at the expense of Pass@k, by avoiding over-reinforcement of already-mastered patterns. The +5.5% improvement over standard OPD confirms that signal *reliability*, not just signal *density*, is a first-order concern.

Stable-OPD (Luo et al., 2026) uncovers a failure mode that neither token weighting nor rollout routing can address: as on-policy training progresses, an implicit feedback loop between the sampling procedure and the distillation objective causes *length inflation*, where rollouts grow progressively longer until they hit the context limit. The truncated trajectories that result introduce systematically biased gradients, triggering a collapse cascade of repetition saturation and sharp performance degradation. This finding has a broader implication for the field: on-policy training creates a closed loop between the student's generation behavior and its training signal, and any bias in one component can self-amplify through the other. Stable-OPD's solution, a reference-based divergence constraint combined with rollout mixture distillation, achieves +7.2% over vanilla OPD by breaking this amplification cycle.

Taken together, these methods reveal a progression from local to global signal management. Token weighting addresses within-sequence heterogeneity, with TIP (Xu et al., 2026b) providing a principled two-axis decomposition that separates informative tokens from noise. Rollout routing addresses across-trajectory heterogeneity, and stability mechanisms address temporal dynamics across the entire training run. Each layer addresses a distinct source of inefficiency, and they compose naturally: one can select tokens by Soft-OR score within rollouts that have been routed by quality within a training loop that is stabilized against length inflation.

4.2.2 *Compute Efficiency*

The signal management strategies above improve *how* training signal is used but do not reduce the cost of *producing* it. The dominant bottleneck of OPD remains the autoregressive generation step: at every training iteration, the student must decode a full sequence token by token before any gradient can be computed. This generation cost, scaling quadratically with sequence length due to KV cache updates, makes OPD 4–5$\times$ more expensive than off-policy SFT (Section 6.4). Two complementary strategies attack this bottleneck from opposite directions: one truncates how much the student generates, the other eliminates the need for a live teacher entirely.

Fast OPD (Zhang et al., 2026a) starts from an empirical observation with far-reaching implications: training signals in on-policy distillation are concentrated in the *prefix* of each output. Even a short teacher-supervised prefix suffices to help the student produce correct answers, because the early tokens in a reasoning chain establish the solution strategy and later tokens mostly execute it mechanically. By truncating both the sampling horizon and the loss computation to this prefix region, Fast OPD matches full OPD performance while reducing training FLOPs by 2$\times$ to 47$\times$. This result suggests that the information-theoretic content of a reasoning trajectory is highly front-loaded, a structural property that future methods could exploit more aggressively.

Lightning OPD (Wu et al., 2026) takes a more radical approach by questioning whether a live teacher is needed at all during on-policy training. The key insight is a *teacher consistency* condition: when the same teacher model is used for both SFT data generation and OPD, the student's rollout distribution remains close to the SFT reference throughout training, meaning precomputed teacher log-probabilities remain valid approximations. Formally, the gradient discrepancy between online and offline OPD is bounded by the χ^2 divergence between the student's current and reference policies. Violating teacher consistency (using different teachers for SFT and OPD) introduces an irreducible gradient bias that causes both variants to converge to suboptimal fixed points. Starting from Qwen3-8B-Base, Lightning OPD reaches 69.9% on AIME 2024 in 30 GPU hours, a 4× speedup over standard OPD.

These two methods reveal a deeper structural insight about on-policy training: the "on-policy" requirement is not binary but admits degrees of approximation. Fast OPD shows that partial rollouts preserve most of the distributional alignment benefit. Lightning OPD shows that stale teacher signals are acceptable when the student's policy drift is bounded. Together, they define a *fidelity–efficiency frontier* that practitioners can navigate based on their compute budget and quality requirements. Speculative KD (Xu et al., 2025c), discussed further in Section 7.2, extends this frontier by drawing on speculative decoding: the student generates candidate tokens that are filtered against the teacher's distribution, with rejected tokens re-sampled from the teacher, producing on-policy training data that balances student exploration with teacher quality.

4.2.3 Curriculum and Difficulty Adaptation

The preceding subsections manage *which signals* to use and *how cheaply* to produce them. A complementary question asks *which problems* the student should tackle at each stage. This question is more subtle than it appears: presenting problems that are too easy wastes compute on already-mastered skills, while problems that are too hard produce uninformative gradients dominated by noise. The optimal training distribution sits in a narrow zone between these extremes, and this zone shifts continuously as the student improves.

PACED (Xu et al., 2026a) provides the most principled treatment of this problem. Through gradient signal-to-noise ratio (SNR) analysis, the authors prove that the distillation gradient SNR vanishes at *both* extremes of student capability: when the student's pass rate approaches 0 (too hard, the gradient is pure noise) or 1 (too easy, the gradient vanishes because there is nothing left to learn). The optimal learning signal concentrates in the "zone of proximal development" where intermediate solve rates produce maximal gradient quality. PACED implements this insight via a Beta kernel weighting scheme $w(p) = p^{\alpha}(1-p)^{\beta}$ that assigns maximum weight to sequences near the student's competence frontier. The framework is provably minimax-robust, with worst-case efficiency loss bounded by $O(\delta^2)$ under multiplicative misspecification of the pass rate estimate. A two-stage divergence schedule (Forward KL for broad mode coverage, then Reverse KL for consolidation) yields the strongest results.

This analysis has a deeper implication that extends beyond curriculum design: it explains *why* on-policy training naturally outperforms off-policy on reasoning tasks. In off-policy settings, the problem difficulty distribution is fixed by the dataset. As the student improves, an increasing fraction of the training data falls into the "too easy" zone where gradients are negligible, leading to diminishing returns. On-policy training, by contrast, automatically adjusts the effective difficulty because the student's rollouts reflect its current capabilities, naturally generating sequences at the boundary of its competence.

AdaSwitch (Peng et al., 2025) addresses the same problem at a finer granularity, operating *within* individual sequences rather than across them. The student begins generating each sequence autonomously, and once its divergence from the teacher exceeds a context-aware threshold, teacher guidance is selectively integrated. This single-switch design preserves semantic coherence (avoiding the jarring token-level alternation of prior mixing methods) while ensuring high-quality supervision at critical junctures. The approach is complementary to PACED: one controls which problems to present, the other controls when to intervene within each problem.

PromptKD (Kim et al., 2024) tackles the difficulty problem from the opposite direction, adapting the *teacher* rather than the student's training data. By prepending learnable prompt tokens to the teacher model, PromptKD causes the teacher to produce "student-friendly" distributions that are more accessible to the student's limited capacity, adding only 0.0007% of the teacher's parameters. This approach implicitly addresses the capacity gap problem: when the teacher's distribution is too complex for the student to match, the standard KL objective drives the student toward a blurred average of the teacher's modes. By simplifying the teacher's output before distillation, PromptKD reduces this gap at its source.

The progression from PACED through AdaSwitch to PromptKD traces an important design spectrum: curriculum can operate at the problem level (which examples), the token level (when to intervene), or the teacher level (how to teach). These approaches are composable and address different bottlenecks. The unifying insight is that the *difficulty gap* between teacher signal and student capability is the primary determinant of gradient quality, and any mechanism that keeps this gap in the productive range, regardless of whether it operates on data, timing, or the teacher itself, will improve training efficiency.

4.3 Representation Alignment across Architectures

The training dynamics and curriculum methods described above assume that teacher and student inhabit the same probability space, sharing a vocabulary over which divergences can be directly computed. In practice, this assumption is increasingly violated. The most compelling teacher-student pairs often span model families (e.g., distilling from Llama into Qwen, or from a text LLM into a vision-language model), where mismatches in tokenizers, hidden dimensions, and positional encodings break the standard $D_{\mathrm{KL}}(p_T \parallel p_\theta)$ computation at its foundation. This architectural mismatch problem is particularly acute in OPD because the student generates its own rollouts using its own tokenizer, and the teacher must then provide supervision over a sequence it would have tokenized differently.

Two complementary strategies address this problem at different levels of abstraction. DSKD (Zhang et al., 2025b) operates in *hidden-state space*, introducing dual projectors that map teacher representations into the student's space and vice versa:

$$\mathcal{L}_{DSKD} = D_{\mathrm{KL}}(P_{T\to S} \parallel P_S) + D_{\mathrm{KL}}(P_{S\to T} \parallel P_T) \tag{22}$$

where $P_{T\to S}$ denotes the teacher's hidden states projected into the student's space and decoded through the student's prediction head. An exact token alignment algorithm handles vocabulary mismatches by identifying identical tokens across tokenizers. Cross-tokenizer distillation (Boizard et al., 2025) instead operates directly in *vocabulary space*, using optimal transport to align logit distributions across different vocabularies:

$$\mathcal{L}_{\mathrm{CrossTok}} = \inf_{\pi\in\Pi(P_S,P_T)} \mathbb{E}_{(z_S,z_T)\sim\pi}\left[\parallel W_{S\to T} z_S - z_T \parallel_2^2\right] \tag{23}$$

The distinction between these approaches highlights a fundamental design trade-off. Hidden-state alignment (DSKD) preserves richer representational structure but requires maintaining two projectors during training, adding memory overhead. Vocabulary-space alignment (cross-tokenizer KD) is more lightweight but operates at a level where information has already been compressed through the vocabulary bottleneck. In either case, the projectors or transport maps must be *co-trained* with the student, because the student's representation space shifts during on-policy training as its policy evolves, a form of representation drift absent in off-policy settings.

Despite these advances, cross-architecture OPD remains substantially harder than same-architecture distillation. Both methods introduce approximation error at the alignment layer, and no current approach handles the case where teacher and student differ simultaneously in tokenizer, hidden dimension, number of layers, and attention structure. The latent-space distillation direction discussed in Section 8 offers a potential path forward by bypassing the vocabulary layer entirely.

4.4 Distillation under Information Constraints

The methods surveyed so far assume white-box access to the teacher's full logit tensor, enabling precise computation of divergences over the teacher's output distribution. This assumption breaks down entirely when the teacher is a proprietary model that exposes neither weights nor vocabulary-wide logits. In these settings, the student cannot compute $D_{\mathrm{KL}}(p_T \parallel p_\theta)$ because p_T is structurally inaccessible. The challenge is therefore to extract as much supervisory signal as possible from limited feedback channels: generated text, scalar scores, or pairwise preferences. Two primary signal construction strategies have emerged, organized below by the type of supervision they recover from these constrained channels.

Adversarial signal construction. When the teacher's internal distribution is inaccessible, an alternative is to *learn* a proxy for it. GAD (Ye et al., 2025) casts black-box distillation as a two-player minimax game. A student generator G produces on-policy responses, while a discriminator D (initialized from the student with a scalar prediction head) distinguishes student outputs from teacher outputs using a Bradley–Terry preference model:

$$\max_G \min_D \; V(G, D) = \mathbb{E}_{(x, y^{\mathcal{T}}) \sim \mathcal{T}} \left[-\log \sigma\Big(D(y^{\mathcal{T}}) - D(G(x)) \Big) \right] \tag{24}$$

The generator maximizes V via REINFORCE, using the discriminator score as an evolving on-policy reward signal. Unlike white-box methods that match full distributions, GAD extracts knowledge through a scalar quality signal, trading distributional fidelity for API compatibility. It achieves consistent gains over SFT baselines, though it inherits the training instability of adversarial objectives, as the discriminator can overfit to stylistic cues rather than semantic quality.

Lion (Jiang et al., 2023) addresses a complementary challenge: even with an adversarial objective, a fixed prompt distribution limits learning to the student's pre-existing weaknesses. Its three-stage adversarial loop, *imitation* (student fine-tuned on teacher responses), *discrimination* (teacher identifies instructions where the student falls short), and *generation* (teacher creates harder instructions targeting identified weaknesses), forms a closed-loop curriculum that progressively narrows the student–teacher gap. Unlike GAD, which uses a learned discriminator, Lion employs the teacher itself as both referee and curriculum designer, eliminating discriminator training at the cost of more teacher API calls. Lion-13B achieves competitive performance on BIG-Bench Hard and AGIEval using only 70k training examples.

Score-based and verbal feedback. Adversarial formulations recover a *relative* signal (better/worse than teacher), but coarser *absolute* feedback can also be surprisingly effective. On-policy Verbal Distillation (OVD) (Xiong et al., 2026) replaces token-level probability matching entirely with trajectory matching using discrete verbal scores (0–9) from teacher models. This eliminates the requirement for token-level alignment between student and teacher vocabularies, dramatically reduces memory consumption by avoiding teacher logit storage, and allows the student to freely explore the output space. OVD delivers up to +12.9% absolute EM improvement on web QA and +25.7% on math benchmarks with a single rollout sample, demonstrating that coarse verbal feedback can outperform dense logit matching when combined with on-policy exploration.

ThinkTuning (RRV et al., 2025) occupies an intermediate position between black-box distillation and RL. Rather than providing scores after the fact, a same-size teacher reviews the student's incorrect answers *during* the GRPO training loop and provides short corrective hints (rather than complete solutions). This interleaving of corrective feedback episodes with RL training acts as an implicit privileged signal that shapes the student's policy gradient updates. ThinkTuning is particularly effective for base models that lack strong prior reasoning behaviors, since these models cannot bootstrap self-reflective behavior through RL alone. Across benchmarks, it achieves +3.85% over zero-shot baselines, with +2.08% on MATH-500, +2.23% on AIME, and +3.99% on GPQA-Diamond over vanilla GRPO.

Preference-based feedback. A third family of black-box methods constructs supervision through preference pairs, leveraging the *relative* quality of competing outputs rather than absolute scores. This approach is theoretically motivated by the observation that humans and teacher models can reliably rank outputs even when they cannot articulate precise quality scores, making preference signals more robust to calibration artifacts than scalar feedback. ORPO-Distill (Singh et al., 2025) formulates cross-architecture distillation as a preference optimization task, contrasting teacher-generated (preferred) and student-generated (dispreferred) traces via an Odds-Ratio Preference Optimization objective. The mixed-policy strategy, combining on-policy student traces with off-policy teacher traces, outperforms both pure off-policy and pure on-policy alternatives, suggesting that the contrastive structure of preference learning (learning what *not* to do alongside what to do) provides complementary signal to the positive-only feedback of score-based methods.

The progression from adversarial (GAD, Lion) to score-based (OVD, ThinkTuning) to preference-based (ORPO-Distill) reflects a spectrum of signal richness versus access requirements. Adversarial methods recover the densest proxy signal but require the most complex training dynamics. Score-based methods are simpler but lose distributional information. Preference-based methods occupy a middle ground, recovering relative quality without requiring calibrated scores, and their connection to DPO-style objectives provides well-understood convergence properties.

Bridging expert knowledge to student distribution. All three families above assume that supervision is generated dynamically during training. A distinct challenge arises when high-quality expert solutions already exist but are *out-of-distribution* for the student (e.g., textbook proofs with implicit reasoning gaps that the student's tokenizer handles differently). Simply fine-tuning on these solutions yields poor transfer because the student never learns to navigate its own reasoning space. DAIL (Mendes et al., 2026) bridges this gap through a two-step process: first transforming expert solutions into detailed, in-distribution reasoning traces, then applying a contrastive objective to focus learning on expert methodologies rather than surface patterns. Using fewer than 1,000 expert solutions, DAIL achieves 10–25% pass@k gains. This result underscores a principle that recurs throughout this survey: the quality of the supervision signal matters less than its *alignment* with the student's current distribution.

4.5 Supervision beyond Distribution Matching

All methods discussed in the preceding subsections share a common assumption: the ideal student is one whose output distribution perfectly mirrors the teacher's. For factual knowledge transfer and instruction following, this assumption is well justified. However, for complex reasoning, code generation, and multi-step planning, perfect distributional matching can be counterproductive. The student may need to discover solution paths that the teacher's default decoding strategy would never produce, or to combine the teacher's structural knowledge with outcome-driven exploration that rewards correct answers regardless of the specific token sequence used to reach them. This subsection covers two complementary directions that move beyond pure distribution matching: chain-of-thought and reasoning transfer, and reward-guided distillation that integrates RL into the distillation pipeline.

4.5.1 *Chain-of-Thought and Reasoning Transfer*

Complex reasoning is where on-policy distillation proves most indispensable, and where off-policy methods encounter their hardest limits. The reason is structural: reasoning is *path-dependent*. A proof that begins "assume for contradiction..." requires entirely different subsequent tokens than one beginning "we proceed by induction..." If the student has never generated the inductive approach during training, it cannot recover at inference time when that path would be optimal. Off-policy training over a fixed dataset of teacher reasoning traces can only cover a finite set of solution paths, while the student needs to navigate its own unique landscape of partial solutions, dead ends, and recovery strategies.

This insight motivates the on-policy formulation for CoT transfer. The student generates its own intermediate reasoning steps $r^S \sim P_\theta(\cdot|x)$, and the teacher evaluates these student-generated chains rather than dictating its own:

$$\mathcal{L}_{\text{CoT-OPD}}(\theta) = \mathbb{E}_{x\sim\mathcal{D}, r^S\sim P_\theta(\cdot|x)} \left[\sum_{t=1}^{|r^S|} D_{\text{KL}}\left(P_\theta(\cdot|x, r^S_{<t}) \parallel P_T(\cdot|x, r^S_{<t}) \right) \right] \tag{25}$$

The Reverse KL direction is mode-seeking by design: rather than forcing the student to spread probability across all teacher-approved continuations (which would blur distinct reasoning strategies), it reinforces the student's own high-probability paths that the teacher validates. This self-consistent learning dynamic, where the student practices its own reasoning style under expert supervision, is why on-policy CoT distillation has become the de facto standard for instilling multi-step reasoning into smaller models (Hsieh et al., 2023).

4.5.2 Reward-Guided Distillation

The integration of RL into the distillation pipeline marks a clear shift in design philosophy: from asking "how closely does the student match the teacher?" to asking "how well does the student *perform*?" This shift is motivated by a key limitation of pure distribution matching. When the teacher's output distribution is suboptimal for the downstream task (e.g., the teacher defaults to verbose reasoning when concise chains would suffice), perfect KL matching reproduces these limitations. Reward-guided methods break this ceiling by introducing an external performance signal that can steer the student toward solutions the teacher would never produce.

The space of KD+RL integration can be organized along a single axis: the *density* of the distillation signal relative to the reward signal. At one extreme, the teacher provides dense token-level logits and the reward provides only sparse outcome feedback. At the other, the teacher provides no direct signal and the reward drives all learning. Most practical methods sit between these extremes.

Dense KD + sparse reward. Frameworks like RLKD (Xu et al., 2025b) exemplify the dense-KD end of the spectrum. RLKD introduces a Generative Structure Reward Model (GSRM) that converts reasoning paths into meta-reasoning-solving steps and computes rewards measuring structural alignment. The objective combines token-level KL regularization with trajectory-level reward:

$$\max_{\theta} \mathcal{J}(\theta) = \mathbb{E}_{x\sim\mathcal{D},y\sim P_\theta(\cdot|x)} \left[R_T(x,y) - \beta D_{\mathrm{KL}}\left(P_\theta(\cdot|x) \parallel P_T(\cdot|x)\right)\right] \tag{26}$$

The mathematical structure of this objective explains *why* the combination works: the KL term provides a dense, analytically computable gradient for token-level imitation, while the reward term contributes a Monte Carlo policy gradient for outcome-driven exploration. The KD component dampens the high variance inherent in policy gradient estimation, while the RL component prevents the student from collapsing onto teacher modes that are suboptimal for the downstream task.

This complementarity enables a phenomenon documented as *Reward Extrapolation* (Yang et al., 2026c): the student discovers novel, structurally sound reasoning paths in regions where the teacher's generation probability $P_T(y|x)$ may be low, yet the outcome reward $R_T(x,y)$ remains highly positive. By optimizing against this joint signal, the student escapes the performance ceiling of strict behavioral cloning, discovering algorithmic shortcuts that the teacher's default decoding strategy would never reach.

The off-policy knowledge problem. A fundamental limitation of pure on-policy RL for reasoning is that the student learns only from its own rollouts, remaining bounded by its initial capabilities. If a weak model cannot solve a problem by any sampling strategy, no amount of on-policy optimization will help. LUFFY (Yan et al., 2025) addresses this by extending GRPO to a mixed-policy objective that incorporates off-policy reasoning traces (e.g., from DeepSeek-R1) alongside the student's on-policy rollouts for joint advantage computation. The critical design challenge is preventing *superficial imitation*: without correction, the student tends to copy high-probability tokens from off-policy traces while ignoring low-probability but critical reasoning steps. LUFFY's policy shaping via regularized importance sampling reweights gradients to emphasize learning from unfamiliar yet effective actions, achieving +6.4 average points over standard RLVR and critically succeeding in training weak foundation models where pure on-policy RL fails entirely.

Joint vs. sequential optimization. A persistent engineering challenge is *how* to combine KD and RL within a training pipeline. Sequential approaches (distill-then-RL or RL-then-distill) suffer from a known instability: the RL stage erases distilled knowledge through reward-driven policy drift, while the KD stage suppresses novel solution paths discovered through exploration. The insight that resolves this tension is that KD and RL objectives need not conflict when they operate on *different components of the gradient signal* (Li et al., 2025).

KDRL (Xu et al., 2025a) operationalizes this by adding an on-policy KL regularizer between student and teacher *during* RL training, directly preventing the forgetting that sequential pipelines suffer. RLAD (Zhang et al., 2026e) refines the approach further with Trust Region Ratio Distillation (TRRD), which replaces the standard teacher-student KL regularizer with a PPO-style likelihood-ratio objective anchored to a teacher-old-policy mixture, yielding advantage-aware, trust-region-bounded distillation on student rollouts. This selective strategy follows the teacher only when its signal improves the policy update and ignores it otherwise, a principled solution to the problem that blind KL minimization can drag the student toward suboptimal trajectories on problems the teacher itself struggles with. The result consistently outperforms both offline distillation and pure GRPO on reasoning benchmarks.

KEPO (Yang et al., 2026b) extends joint optimization to the vision-language domain, where a distinct failure mode emerges: sparse RLVR fails due to exploration collapse when initial solve rates are near zero. By conditioning preference optimization on teacher knowledge, KEPO provides the dense signal necessary to bootstrap capable policies, a finding that suggests reward-guided OPD is most valuable precisely when pure RL struggles most.

The evolution from off-policy reasoning distillation (DeepSeek-R1 style) to on-policy methods reveals a clear progression. Off-policy SFT on teacher reasoning traces establishes a strong foundation by transferring reasoning *structure*. On-policy methods then close the remaining gap by addressing the distribution mismatch between the teacher's reasoning style and the student's evolving capabilities. OPSD (Zhao et al., 2026b) and PACED (Xu et al., 2026a) represent this next step, applying genuine on-policy distillation where the student generates its own reasoning rollouts and the teacher provides token-level supervision on those student-generated sequences. An orthogonal but practically important direction is reasoning compression: CRISP (Sang et al., 2026) applies on-policy self-distillation to compress verbose chains of thought, reducing token count by 57–59% on MATH-500 while *improving* accuracy, revealing that much of the verbosity in distilled reasoning models is a trainable inefficiency rather than a necessary cost of multi-step reasoning.

Targeted correction vs. wholesale imitation. The reward-guided methods above still provide uniform supervision across the entire trajectory. A distinct failure mode emerges from this uniformity: distribution-matching treats all reasoning tokens as equally important, providing dense supervision even on correctly solved portions. When the student makes a single error in step 3 of a 10-step proof, wholesale imitation retrains all 10 steps, diluting the corrective gradient. Methods in this group make the teacher's signal *targeted* rather than exhaustive.

SuperCorrect (Yang et al., 2025b) introduces a two-stage framework: hierarchical thought templates scaffold the student's reasoning at both strategic and step-level granularity, then cross-model DPO teaches the student to locate and resolve its own errors:

$$\mathcal{L}_{\text{SuperCorrect}}(\theta) = \mathcal{L}_{\text{template}}(\theta) + \lambda \cdot \mathcal{L}_{\text{cross-DPO}}(\theta) \tag{27}$$

SCoRe (Lyu et al., 2025) pushes the targeting principle to its logical extreme: rather than providing complete correction traces, the teacher identifies and corrects *only the earliest error* in the student's trajectory, concentrating all supervision at the single point of maximum leverage. This approach exploits a key structural property of reasoning chains, namely that errors cascade, so correcting the root cause eliminates all downstream consequences simultaneously. The contrast between SuperCorrect's template-based and SCoRe's error-first approaches reveals two viable strategies for the same problem: top-down structural scaffolding versus bottom-up error triage.

From signal matching to environment reconstruction. All methods discussed so far treat the teacher as an external signal provider, whether that signal is logits, rewards, corrections, or preferences. A more radical question asks whether the student can recover the teacher's *underlying optimization landscape* and learn by re-experiencing it. $\mathcal{X}$-KD (Cai & Yuan, 2026) adopts the AVRIL framework to jointly model the teacher's latent reward function and perform policy distillation, encouraging the student to optimize in a reconstructed version of the teacher's training environment rather than simply mimicking its outputs. TSD-KD (Kim & Baek, 2026) takes a complementary approach by combining indirect distillation

Table 2: Comprehensive comparison of white-box on-policy distillation methods.

Method	Year	Granularity	Divergence	Access	On-Policy Formulation	Key Innovation
GKD (Agarwal et al., 2024)	2024	Token	F-KL / R-KL / JSD	White-box	Student Rollouts	Unified OPD framework
G-OPD (Yang et al., 2026c)	2026	Token	KL-Constrained RL	White-box	Reward Extrapolation	OPD as RL, student surpasses teacher
MiniLLM (Gu et al., 2024)	2024	Seq.	Reverse KL	White-box	REINFORCE	Sequence-level reward
DistiLLM (Ko et al., 2024)	2024	Token	Skew KL (SKL+SRKL)	White-box	Interpolated	α-smoothing for KL
DistiLLM-2 (Ko et al., 2025)	2025	Token	Contrastive SKL/SRKL	White-box	Batched On-Policy	Asymmetric loss per data type
DSKD (Zhang et al., 2025b)	2025	Token	Dual-Space KL	White-box	Cross-Space Projection	Vocabulary-agnostic KD
Constrained (Zimmer et al., 2025)	2025	Seq.	Constrained KL	White-box	State-Augmented CMDP	Reward-aware distillation with hard KL constraint
KETCHUP (Fan et al., 2025)	2025	Seq.	Reverse KL	White-box	K-step Return	Multi-step REINFORCE with reduced variance
Entropy-Aware (Jin et al., 2026)	2026	Token	Adaptive FKL+RKL	White-box	Entropy-Adaptive	High-entropy robustness
Fast OPD (Zhang et al., 2026a)	2026	Hybrid	R-KL on Prefix	White-box	Prefix-Truncated	Reasoning prefix reuse
PACED (Xu et al., 2026a)	2026	Hybrid	Adaptive	White-box	Frontier Curriculum	Student competence boundary
$\mathcal{X}$-KD (Cai & Yuan, 2026)	2026	Hybrid	AVRIL + KL	White-box	Reward-Guided	Teacher environment reconstruction
AdaSwitch (Peng et al., 2025)	2025	Hybrid	KL (adaptive)	White-box	One-time Switch	Adaptive on/off-policy switching
REOPOLD (Ko et al., 2026)	2026	Token	Reward-clipped R-KL	White-box	Relaxed Imitation	OPD as policy optimization
TSD-KD (Kim & Baek, 2026)	2026	Token	Dual KD (indirect + direct)	White-box	Student Re-ranking	Token-selective student-centric
Veto (Jang et al., 2026)	2026	Token	Geometric Bridge KL	White-box	Adaptive Gradient Veto	Logit-space bridge for stable OPD
DASD (Yan et al., 2026)	2026	Seq.	Distribution-aligned	White-box	On-policy Correction	Distribution alignment for seq. KD
DDT (Zhang et al., 2026b)	2026	Seq.	IDFT + Hinted Decoding	White-box	Distribution-aligned SFT	On-policy SFT theory
Lightning OPD (Wu et al., 2026)	2026	Token	Offline R-KL	White-box	Offline On-Policy	Teacher consistency, 4$\times$ speedup
SCOPE (Zheng et al., 2026)	2026	Hybrid	Dual-path R-KL + MLE	White-box	Correctness-Routed	Perplexity-weighted signal calibration
TIP (Xu et al., 2026b)	2026	Token	Soft-OR weighted R-KL	White-box	Two-axis Token Selection	Entropy+divergence blind-spot recovery

Table 3: Experimental configurations of representative on-policy distillation methods. "Self" indicates self-distillation where the same model serves as both teacher and student. Dashes indicate information not available from the original paper.

Method	Teacher	Student	Loss / Objective	Primary Tasks
White-Box Token-Level				
GKD (Agarwal et al., 2024)	T5-XL (3B)	T5-Small (77M) / Base (250M) / Large (800M)	F-KL / R-KL / JSD	XSum, WMT, GSM8K
G-OPD (Yang et al., 2026c)	Qwen3-4B-RL / Qwen3-30B-A3B	Qwen3-4B-Non-Thinking	KL-constrained RL (λ=1.25)	AIME, MATH, HumanEval
DistiLLM (Ko et al., 2024)	GPT-2 XL (1.5B), OpenLLaMA-7B	GPT-2 (124M), OpenLLaMA-3B	Skew KL (SKL+SRKL, α=0.1)	Dolly, S-NI, XSum
DistiLLM-2 (Ko et al., 2025)	Gemma-2-9B-Inst, Mistral-7B-Inst	Gemma-2-2B, Danube2-1.8B	Contrastive SKL/SRKL	AlpacaEval, GSM8K, HumanEval
Entropy-Aware (Jin et al., 2026)	Qwen3-8B	Qwen3-0.6B/1.7B/4B-Base	Entropy-gated R-KL/F-KL	AIME, MATH, MMLU
Veto (Jang et al., 2026)	Qwen2-7B-IT	Qwen2-0.5B-IT	Geometric bridge KL	Math, Code, Summarization
Revisiting OPD (Fu et al., 2026)	OpenThinker3-7B	Qwen2.5-7B-IT	Truncated R-KL + top-p	MATH, Agentic + Math
DSKD (Zhang et al., 2025b)	Qwen2.5-7B-Inst, Qwen2.5-Math-7B	Qwen2.5-1.5B, Llama-3.2-1B	Dual-Space KL	Instruction following, Reasoning, Code
REOPOLD (Ko et al., 2026)	SkyWork-OR1-7B/32B	DeepSeek-R1-Distill-Qwen-1.5B/7B	Reward-clipped R-KL	AIME, Math, Visual reasoning
TSD-KD (Kim & Baek, 2026)	Qwen2.5-14B-Inst	Qwen2.5-1.5B-Inst	Token-selective dual KD	Math, Code, Science
White-Box Sequence-Level & Hybrid				
MiniLLM (Gu et al., 2024)	GPT-2 1.5B, GPT-J 6B, OPT-13B	GPT-2 125M–760M, GPT-Neo 2.7B, OPT 1.3B–6.7B	Seq-level R-KL (REINFORCE)	Instruction following
Constrained (Zimmer et al., 2025)	Qwen2.5-7B-Math	Qwen2.5-1.5B-Math	CMDP (reward + KL constraint)	MATH, GSM8K
KETCHUP (Fan et al., 2025)	FLAN-T5-XL (3B)	T5-Base	K-step return REINFORCE	XSum, Europarl, GSM8K
Fast OPD (Zhang et al., 2026a)	Qwen3-8B	Qwen3-1.7B / 8B-Base	R-KL on prefix only	AIME, MATH, LiveCode
PACED (Xu et al., 2026a)	Qwen3-14B	Qwen3-8B	Beta-kernel weighted KL	AIME, MATH
AdaSwitch (Peng et al., 2025)	Qwen2.5-3B, Llama-3.1-3B, Gemma-7B	Qwen2.5-0.5B, Llama-3.1-1B, Gemma-2B	Adaptive on/off-policy KL	Summarization, GSM8K, GSM-Plus
$\mathcal{X}$-KD (Cai & Yuan, 2026)	T5-Large (780M)	T5-Small (77M) / Base (250M)	AVRIL-based KL	Summarization, Translation, Arithmetic
DASD (Yan et al., 2026)	gpt-oss-120b / Qwen3-Next-80B-A3B	Qwen3-4B-Instruct (DASD-4B)	Distribution-aligned seq. KD	AIME, MATH, LiveCode
DDT (Zhang et al., 2026b)	N/A (on-policy SFT)	Qwen2.5-7B-Inst	IDFT + Hinted Decoding	Alignment, Reasoning
Lightning OPD (Wu et al., 2026)	Qwen3-8B, Qwen3-32B	Qwen3-4B-Base, Qwen3-8B-Base	Offline R-KL (precomputed teacher logprob)	AIME, HMMT, LiveCode
SCOPE (Zheng et al., 2026)	SkyWork-OR1-Math-7B, Qwen3-8B-Inst	R1-Distill-Qwen-1.5B, Qwen3-1.7B-Base	Dual-path perplexity-weighted R-KL + MLE	AIME, MATH-500, Minerva, Olympiad
TIP (Xu et al., 2026b)	Qwen3-8B, Llama-70B-Inst, Qwen2.5-14B-Inst	Qwen3-4B, Llama-3.1-8B-Inst, Qwen2.5-1.5B-Inst	Soft-OR token-selected R-KL	MATH-500, AIME, DeepPlanning
Black-Box				
GAD (Ye et al., 2025)	GPT-5-Chat (black-box)	Qwen2.5-14B-Inst	Adversarial (minimax)	LMSYS-Chat, Dolly, Vicuna
Lion (Jiang et al., 2023)	ChatGPT (black-box)	LLaMA-7B / 13B	Adversarial curriculum	BIG-Bench Hard, AGIEval
OVD (Xiong et al., 2026)	QwQ-32B / Environment Agent	Qwen2.5-3B, Llama-3.2-3B, Qwen2.5-Math-1.5B	Verbal scores (0–9)	Web QA, MATH
ORPO-Distill (Singh et al., 2025)	InternLM2.5-7B-Chat	InternLM2.5-1.8B, TinyLlama-1.1B	ORPO (reference-free)	GSM8K, ARC, MedQA, StrategyQA
DAIL (Mendes et al., 2026)	Expert solutions (black-box)	Qwen2.5-7B-Inst, Qwen3-8B	Contrastive imitation	AIME, GPQA
Self-Play & Self-Distillation				
SPIN (Chen et al., 2024)	Self (iter $t-1$)	Zephyr-7B-SFT	Self-play DPO	Open LLM, MT-Bench, BBH
OPSD (Zhao et al., 2026b)	Self (privileged info-conditioned)	Qwen3-1.7B/4B/8B-Inst	F-KL / JSD on rollouts	AIME, MATH, HMMT
CRISP (Sang et al., 2026)	Self (concise prompt)	Qwen3-8B / 14B	Per-token R-KL	AIME, MATH-500
GATES (Stein et al., 2026)	Self (document-conditioned)	Qwen3-4B-Base	Consensus-gated KL	Open-domain QA
SDPO (Hübotter et al., 2026)	Self (textual feedback)	Qwen3-8B, Qwen2.5-Inst	Self-distill + credit assign	Code, math
MTP Self-Distill (Kirchenbauer et al., 2026)	Self (same checkpoint)	Llama-3.1-8B, Qwen2.5-7B	Multi-token prediction	GSM8K, MATH, MMLU
OPCD (Ye et al., 2026b)	Self (context-augmented)	Qwen3-1.7B / 4B / 8B	R-KL context distillation	Knowledge, sys prompts
OEL (Ye et al., 2026a)	Self (experiential)	Qwen3-1.7B / 4B / 8B	KL + context distillation	Text-based games
SDFT (Shenfeld et al., 2026)	Self (demo-conditioned)	Qwen2.5-7B-Inst	Self-distill + KL reg	Skill learning, Tool Use
SD-ZERO (He et al., 2026)	Self (revision-conditioned)	Qwen3-4B-Inst, OLMo-3-7B-Inst	On-policy self-distill via revision	AIME, MATH, Codeforces, LCB
Priv. Info. Distill (Penaloza et al., 2026)	Self (PI-conditioned)	Qwen3-4B / 8B, R1-Distill-Llama-8B	Privileged self-distill KL	Long-horizon RL
HDPO (Ding, 2026)	Self (privileged, ground-truth)	Qwen2.5-Math-1.5B-Inst	GRPO + JSD self-distill	MATH, AMC, AIME
RLSD (Yang et al., 2026a)	Self (revision-conditioned) + RLVR	Qwen3-VL-8B-Inst	Self-distill magnitude + RLVR direction	MATH, VQA benchmarks
SRPO (Li et al., 2026b)	Self (SDPO-guided) + GRPO	Qwen3-8B	Sample-routed self-distill + RL	MATH, AIME, code
SSB (Mitra & Ulukus, 2025)	Self (correct/incorrect rollouts)	Qwen2.5-3B-Inst	Semantic soft bootstrapping	MATH-500, AIME2024
RLTF (Song et al., 2026)	Self (critique-conditioned)	Qwen3-8B	Self-distill via text feedback	Math, code, creative writing
On-Policy SFT (Zhao et al., 2026a)	Self (correct+concise rollouts)	DeepSeek-R1-1.5B	On-policy SFT (simplified RL)	MATH, AIME, code
SSD (Zhang et al., 2026c)	Self (temperature-sampled rollouts)	Qwen3-30B-Inst, Llama-3.1-8B	On-policy SFT (temp. diversity)	LiveCodeBench v6
π-Play (Zhang et al., 2026d)	Self (QCP-conditioned teacher)	Qwen3-4B/4B-Inst/8B	Self-distill via QCP privileged info	NQ, TriviaQA, HotpotQA, MuSiQue, 2WikiMQA
ThinkTuning (RRV et al., 2025)	Same-size teacher (corrective hints)	Llama-3.2-3B-Instruct	GRPO + teacher corrective feedback	MATH, AIME, GPQA-Diamond
Stable-OPD (Luo et al., 2026)	Larger teacher	Qwen2.5-Math-1.5B/7B	R-KL + reference divergence + rollout mix	MATH, AIME
Reasoning & RL Integration				
RLKD (Xu et al., 2025b)	DeepSeek-R1 paths + GSRM (7B)	DeepSeek-R1-Distill-Qwen-7B	KL-regularized RL + GSRM	Math reasoning
RLAD (Zhang et al., 2026e)	Qwen3-8B / 32B	Qwen3-0.6B / 1.7B / 8B	Trust-region ratio distill	MATH, GSM8K, AIME
AlignDistil (Zhang et al., 2025a)	Synthetic DPO + reverse DPO	Qwen2-1.5B-Inst, Qwen2.5-1.5B-Inst	DPO + token-level KD	AlpacaEval, MT-Bench, Arena-Hard
LUFFY (Yan et al., 2025)	DeepSeek-R1 traces (black-box)	Qwen2.5-Math-7B / 1.5B	Mixed-policy GRPO	AIME, AMC, MATH-500, Minerva
SuperCorrect (Yang et al., 2025b)	o1-mini, gpt-4o-mini (black-box)	DeepSeekMath-7B, Qwen2.5-Math-7B, Llama-3-8B	KD + self-correction reward	MATH, GSM8K
KDRL (Xu et al., 2025a)	Skywork-OR1-Math-7B	R1-Distill-Qwen-1.5B	Joint KD + RL objective	AIME, MATH
KEPO (Yang et al., 2026b)	Qwen3-VL-32B	Qwen3-VL-2B	KD + preference optim.	OmniMedVQA
SCoRe (Lyu et al., 2025)	Qwen2.5-72B-Inst	Qwen2.5-7B/3B-Inst, Llama-3.1-8B	Earliest-error correction	AIME, MATH, Multi-hop QA
Multimodal & Domain-Specific				
VOLD (Bousselham et al., 2025)	Qwen3-8B (text-only)	Qwen2.5-VL-3B-Inst	GRPO + token-level KL	MMMU-Pro, MathVision
Video-OPD (Li et al., 2026c)	Qwen3-VL-32B-GRPO	Qwen3-VL-8B-Inst	On-policy temporal distill	TVG (Charades, ActivityNet)
X-OPD (Cao et al., 2026)	Qwen3-Omni (text mode)	Qwen3-Omni-A3B (speech mode)	Cross-modal on-policy KL	Speech understanding
CORD (Hu et al., 2026)	Text-mode LALM (same model)	Audio-mode LALM	Cross-modal self-distill (weighted R-KL + GRPO)	Audio QA/reasoning
VLA-OPD (Zhong et al., 2026)	Expert policy (demonstrations)	VLA student	Reverse KL on self-generated trajectories	LIBERO, RoboTwin2.0
HY-Embodied-0.5 (Tencent Robotics X and HY Vision Team, 2026)	32B expert VLM	MoT-2B embodied VLM	Multi-stage OPD	Embodied VQA, spatial reasoning
Systems & Scaling				
Speculative KD (Xu et al., 2025c)	Gemma-7B-IT, Qwen2-7B-IT	Gemma-2B, Qwen2-0.5B	Block-verified on-policy KL	Translation, Dialogue
DistillSpec (Zhou et al., 2023)	T5-XL (3B), GPT-like (234M)	T5-Small (77M), GPT-like (33M)	On-policy draft $\rightarrow$ target	XSum, CNN/DM
Cross-Tok. KD (Boizard et al., 2025)	Llama-2-7B-Chat, Mistral-7B-Inst	OPT-350M, Pythia-160M–1B, Bloomz-560M	Latent OT alignment	QA, Summarization
Gemma 2 (Gemma Team et al., 2024)	Gemma 2 27B $\rightarrow$ 9B $\rightarrow$ 2B	Gemma 2 2B / 9B / 27B	KD in pre-training	MMLU, HellaSwag, GSM8K
Qwen3 (Yang et al., 2025a)	Qwen3-32B / 235B-A22B	Qwen3-0.6B–14B, 30B-A3B	On-Policy Distillation	AIME, MATH, LiveCode
MiMo-V2 (Xiaomi LLM-Core Team et al., 2026)	Domain-specialized teachers (RL/SFT)	MiMo-V2-Flash (309B/15B MoE)	Multi-teacher logit + reward	AIME, MATH, GPQA
KAT-Coder-V2 (Li et al., 2026a)	5 domain specialists (RL/SFT)	Unified agentic coder	Specialize-then-Unify OPD	SWE-bench, WebCoding
Nemotron-Cascade 2 (Yang et al., 2026d)	Domain RL/SFT teachers	Nemotron 30B MoE (3B active)	Cascade RL + domain OPD	IMO, IOI, ICPC
OpenClaw-RL (Wang et al., 2026b)	Self (hindsight-guided OPD)	Qwen3-4B	PRM + Hindsight OPD	Personal/Terminal/GUI/SWE
DP-OPD (Khadem et al., 2026)	GPT-2 Large 774M (frozen)	DistilGPT-2 82M	GKD + DP-SGD (student only)	Yelp, BigPatent
OPD-AV (Afsharrad et al., 2026)	Qwen3-8B (SFT)	Qwen3-1.7B	GKD (5$\times$ compression)	nuScenes
Skill-SD (Wang et al., 2026a)	Self (skill-conditioned teacher)	Qwen3-4B-Instruct	IW Reverse KL + GRPO	AppWorld, Sokoban
ORBIT (Liang et al., 2026)	Multi-teacher (mode-specific RL experts)	DeepSeek-R1-Distill-1.5B, Qwen3-4B, Nemotron-7B	Multi-teacher R-KL mode fusion	AIME, GPQA, MMLU-Pro

(student generates candidates that the teacher re-ranks via preference feedback) with direct distillation (KL matching applied selectively to high-entropy tokens where teacher-student gaps are largest). These methods represent the current frontier of reward-guided OPD: moving from matching the teacher's *behavior* to understanding its *incentives*.

5 Self-Distillation

The teacher-guided methods of the preceding section assume access to a separate, typically larger model that provides supervisory signals during training. In many practical scenarios, however, such access is unavailable: the teacher may be proprietary without API access, prohibitively expensive to query at the scale required for on-policy training, or simply nonexistent for a novel domain. Self-distillation addresses this constraint by having the model generate its own distillation targets. Rather than relying on an external oracle, the model exploits asymmetries within itself, between its current and past checkpoints, between its behavior with and without privileged information, or between its verbose and concise reasoning modes, to construct a training signal that drives continued improvement.

The three subsections below trace a natural progression from the richest to the most minimal internal signal. Privileged-information methods (Section 5.1) exploit the strongest form of internal asymmetry: training-time access to factual ground truth, source documents, or behavioral preferences that are withheld at inference. Self-play methods (Section 5.2) reduce the signal to the most minimal form, relying entirely on the model's own current or historical outputs with no additional information of any kind, accepting the saturation risk this entails. Finally, external-feedback methods (Section 5.3) resolve the saturation problem by reconnecting the model to objective reality, incorporating signals from the environment such as compiler outputs, test results, or textual critiques, that originate outside the model itself. This arc from information richness through minimalism and back to environmental grounding reflects the fundamental tension in self-distillation: the methods most resistant to saturation are precisely those most dependent on external signals, while the methods that need no external signals are the most prone to it.

5.1 Privileged Information

A model can serve as its own teacher when it has access to information at training time that is unavailable at inference. This *privileged information* (PI), whether ground-truth answers, source documents, or conciseness instructions, creates an asymmetry within a single model that drives self-improvement without any external teacher. The central insight is that conditioning the same model on additional context produces a strictly stronger policy that can supervise the unconditioned version, turning a single network into both teacher and student. The progression of methods in this area traces an expanding arc of what counts as "privileged": starting from direct ground-truth labels, relaxing to indirect contextual cues, then extending to deployment-time behavioral preferences, and finally applying PI to target reasoning efficiency itself. Each step expands the practical reach of self-distillation while introducing new challenges around supervision reliability.

Ground-truth as privileged information. The most direct instantiation conditions the model on the correct answer during training. On-Policy Self-Distillation (OPSD) (Zhao et al., 2026b) constructs the teacher policy by conditioning on both the problem x and the ground-truth answer $y^\star$, while the student conditions only on x. The student generates on-policy rollouts, and the teacher provides dense token-level supervision on these student-generated sequences:

$$\mathcal{L}_{\text{OPSD}}(\theta) = \mathbb{E}_{(x,y^\star)\sim\mathcal{S}}\mathbb{E}_{\hat{y}\sim p_\theta(\cdot|x)}\left[\sum_{n=1}^{|\hat{y}|} D\big(p_T(\cdot|x, y^\star, \hat{y}_{<n}) \parallel p_\theta(\cdot|x, \hat{y}_{<n})\big)\right] \tag{28}$$

On competition-level math benchmarks (AIME 2024/2025, HMMT), OPSD matches or exceeds GRPO at 4B and 8B scales while using only a single rollout per problem versus GRPO's eight, though at 1.7B scale the gains over GRPO are minimal, indicating that sufficient model capacity is necessary for the PI mechanism to provide useful supervision. A complementary limitation arises when all rollouts fail (the "cliff prompt" problem), causing the policy gradient to vanish entirely. This failure mode directly parallels PACED's finding that gradient SNR vanishes when student pass rate approaches zero (Section 4.2), but with an important difference: PACED addresses this through curriculum selection (avoiding problematic prompts), while HDPO (Ding, 2026) addresses it through signal augmentation. HDPO augments GRPO with a JSD-based self-distillation term on ground-truth-conditioned rollouts, ensuring non-zero gradients even at the boundary of the student's competence, directly patching the cliff-prompt vulnerability without restricting the problem distribution.

Relaxing the ground-truth requirement. Ground-truth labels are not always available, and this scarcity motivates the next generalization. GATES (Stein et al., 2026) demonstrates that *any* training-time context can serve as PI by having a single model act as *tutor* (conditioned on a source document) and *student* (answering from the question alone). This is a non-trivial relaxation: unlike OPSD's ground-truth conditioning, a source document does not guarantee that the tutor will produce the correct answer, so a consensus-based gating mechanism suppresses the gradient when tutor uncertainty is high, preventing the model from reinforcing unreliable self-supervision. The gating mechanism in GATES thus serves the same functional role as SCOPE's rollout routing in the teacher-guided setting (Section 4.2): both identify unreliable supervision and attenuate it rather than applying it uniformly. Penaloza et al. (2026) generalize the OPSD and GATES paradigms into a unified theoretical framework, formalizing the conditions under which privileged self-distillation provably improves over standard training, and showing that the benefit of PI grows with task horizon length.

From training-time PI to deployment-time context. The PI paradigm extends naturally beyond mathematical reasoning to any setting where the model has asymmetric access to useful context. On-Policy Context Distillation (OPCD) (Ye et al., 2026b) treats system prompts, exemplars, and retrieval context as privileged information, enabling a model to internalize behaviors that would otherwise require expensive in-context demonstrations at inference time. This is a qualitative shift in the PI concept: rather than conditioning on factual ground-truth, the model conditions on *behavioral instructions*, learning to produce system-prompt-consistent outputs even when the prompt is absent at test time. Building on this foundation, Online Experiential Learning (OEL) (Ye et al., 2026a) extends to continuous post-deployment learning, using interaction outcomes as dynamic PI that updates the model's teacher policy as it accumulates experience. OEL thus closes a loop that OPCD opened: OPCD distills behavioral priors from static context, while OEL refreshes these priors from live experience, together enabling a model that improves autonomously without any external teacher.

Targeting reasoning efficiency. A particularly impactful application of PI is reasoning compression. CRISP (Sang et al., 2026) demonstrates that a model prompted to "be concise" can serve as a teacher for its own verbose version, reducing chain-of-thought token count by 57–59% on MATH-500 while *improving* accuracy by 9–16 percentage points. This result reveals a surprising insight that connects directly to the self-play saturation analysis of Section 5.3: much of the verbosity in distilled reasoning models is a trainable inefficiency, not a necessary feature of competent reasoning. Where self-play methods often struggle to improve because the model's verbose rollouts reinforce their own style, CRISP's PI-based approach sidesteps this saturation by providing an explicit reference policy (the concise version) that differs structurally from the student's default. The efficiency of the resulting models further demonstrates that PI-based self-distillation is not just a substitute for teacher-guided distillation but can, in specific dimensions, surpass it.

5.2 Self-Play

The methods above exploit informational asymmetries to create an internal teacher–student gap. A more radical question is whether a model can improve using nothing but its own rollouts, with no privileged information, no external feedback, and no additional context of any kind. The answer, as demonstrated by a growing family of self-play methods, is a qualified yes, subject to important limitations that motivate the breakthrough strategies discussed in Section 5.3.

Self-play as a two-player game. The foundational approach casts self-improvement as an adversarial game between successive checkpoints. SPIN (Chen et al., 2024) trains the updated model $p_{\theta_{t+1}}$ to distinguish the previous iteration's generations from human-written responses:

$$\mathcal{L}_{\text{SPIN}} = \mathbb{E}_{x,y\sim p_{\text{data}},y'\sim p_{\theta_t}}\left[\ell\left(\lambda\left(\log\frac{p_{\theta_{t+1}}(y|x)}{p_{\theta_t}(y|x)} - \log\frac{p_{\theta_{t+1}}(y'|x)}{p_{\theta_t}(y'|x)}\right)\right)\right] \tag{29}$$

SPIN provides a theoretical convergence guarantee: the global optimum is achieved when $p_\theta = p_{\text{data}}$. Starting from Zephyr-7B-SFT, it improves MT-Bench from 5.94 to 6.78 over 3 iterations, with diminishing returns at each round. This rapid saturation, analyzed in Section 5.3, reveals a fundamental limitation of pure self-play: it recycles existing knowledge rather than generating new knowledge.

Temporal self-supervision. If iterative self-play between adjacent checkpoints saturates quickly, a natural question is whether a model's own training history can supply a richer supervisory signal. Self-Distillation Fine-Tuning (SDFT) (Shenfeld et al., 2026) addresses this in the continual learning setting, positioning on-policy self-distillation as a mechanism for forgetting resistance via implicit KL regularization against the model's own prior state. By distilling the current policy toward its earlier checkpoint, SDFT maintains distributional proximity to previously learned behaviors while still absorbing new task knowledge, effectively converting the model's own optimization trajectory into a regularization resource.

Self-play for efficiency. Beyond improving accuracy, self-play can target inference speed and reasoning efficiency, suggesting that the same mechanism that generates improved solutions can also generate *faster* solutions. Multi-Token Prediction (MTP) via self-distillation (Kirchenbauer et al., 2026) converts a pre-trained autoregressive model into a fast multi-token predictor, achieving more than 3× faster decoding on GSM8K at less than 5% accuracy drop. On-Policy SFT (Zhao et al., 2026a) challenges the necessity of RL for reasoning compression entirely, showing that a truncation-based length penalty simplifies the RL optimization to standard SFT on self-generated data filtered for correctness and conciseness. This result is significant because it suggests that much of what RL contributes to reasoning efficiency can be recovered by the simpler mechanism of selective self-distillation: generate many solutions, keep the short correct ones, train on those.

Minimalist self-distillation. The extreme end of the simplicity spectrum is Simple Self-Distillation (SSD) (Zhang et al., 2026c), which requires no RL, no verifier, no external teacher, and no code execution. SSD samples solutions at a training-time temperature, fine-tunes via standard SFT, and reshapes token distributions in a context-dependent manner. On LiveCodeBench v6, SSD improves Qwen3-30B-Instruct from 42.4% to 55.3% pass@1. This result, together with On-Policy SFT's findings, reveals a deeper principle: when the base model is sufficiently capable, even the simplest form of self-play (filtering one's own diverse outputs for quality) yields substantial gains. The critical requirement is not a sophisticated algorithm but sufficient *diversity* in the model's sampling distribution to produce high-quality variants that can serve as training targets. However, this conclusion also reveals the fundamental boundary of pure self-play: when diversity is achieved through temperature sampling alone, the model's improvement is ultimately bounded by the information already present in its pre-training distribution. Bridging this boundary requires connecting the model to external reality, which motivates the external-feedback methods that follow.

Self-play with privileged self-distillation. The methods above either use privileged information (Section 5.1) or self-play, treating them as distinct paradigms. π-Play (Zhang et al., 2026d) unifies these two approaches within a single multi-agent self-evolution framework for deep search agents. The key observation is that self-play naturally produces a byproduct that has been overlooked: during task generation, the examiner agent constructs a *question construction path* (QCP) that captures the reverse solution process, and this QCP provides high-quality privileged context for the teacher model at no additional cost. This converts the conventional sparse-reward self-play loop into a dense-feedback self-evolution system in which three co-evolving agents, an examiner π^E_ϕ, a teacher π^T_ψ conditioned on the QCP as privileged context, and a student π^S_θ trained without it, are jointly optimized through alternating updates. As the student strengthens, the examiner generates harder tasks, and the teacher's behavior is softly constrained to track the student via a KL penalty, establishing a self-adjusting curriculum that requires no external data, no human feedback, and no curated privileged information. On the Qwen3 series (4B and 8B), data-free π-Play surpasses fully supervised search agents (Search-R1 by 5–15%) and improves evolutionary efficiency by 2–3$\times$ over conventional self-play (Dr.Zero), with the largest gains on multi-hop benchmarks where the teacher's token-level credit assignment provides the most leverage. π-Play thus demonstrates that the self-play saturation problem analyzed in Section 5.3 can be mitigated without external feedback by *internally generating* privileged information through the self-play process itself, a mechanism orthogonal to the external-verification and distillation-RL strategies discussed below.

5.3 External Feedback

The self-play methods above rely entirely on the model's own logit distributions, which can become noisy or degenerate after extended training. External signals, such as compiler outputs, test results, and textual critiques, offer a richer supervision source that grounds self-distillation in objective reality and bridges the gap to reinforcement learning. The key design question is how to convert sparse, binary, or unstructured environmental feedback into a dense training signal compatible with token-level optimization.

From binary rewards to dense supervision. The simplest external signal is a binary correctness indicator (pass/fail from a verifier), but naively using it as a reward yields the sparse credit-assignment problem endemic to RL. Two complementary strategies have emerged for densifying this signal. SD-ZERO (He et al., 2026) introduces a dual-role architecture where a single model plays both Generator and Reviser: the Generator produces on-policy responses, and the Reviser, conditioned on the Generator's output and its binary correctness signal, produces an improved version. The Generator then distills from the Reviser's token-level distribution, converting binary feedback into dense self-supervision. On Qwen3-4B-Instruct, SD-ZERO achieves 68.3% on AIME 2024, outperforming GRPO (62.5%). Self-Distilled RLVR (RLSD) (Yang et al., 2026a) takes a different decomposition, using self-distillation to determine the *magnitude* of per-token policy updates while environmental RLVR feedback anchors the *direction*, combining the stability of distillation with the grounding of external verification. At 200 training steps, RLSD surpasses GRPO trained for 400 steps on Qwen3-VL-8B-Instruct.

Structured textual feedback. Beyond binary rewards, richer textual feedback enables more targeted learning by providing not just *that* a response failed but *why*. SDPO (Hübotter et al., 2026) replaces scalar rewards with structured feedback from runtime errors, failing unit tests, and LLM judge evaluations, constructing fine-grained credit assignment that attributes success or failure to specific reasoning steps rather than entire trajectories. RL from Text Feedback (RLTF) (Song et al., 2026) extends this idea by using natural language critiques from an automated judge, training the model's single-turn policy to match its own feedback-conditioned second-turn generations. The progression from SD-ZERO's binary signal through SDPO's structured errors to RLTF's free-form critiques traces a clear trajectory: as the external feedback becomes richer, the self-distillation objective becomes more precisely targeted, enabling the model to correct specific failure modes rather than globally adjusting its policy.

Hybrid routing: combining strategies by sample quality. Rather than applying a single strategy uniformly, the methods above can be composed by routing different samples to different training objectives based on their quality. Sample-Routed Policy Optimization (SRPO) (Li et al., 2026b) demonstrates this principle by routing correct samples to GRPO's reinforcement path (reward-weighted likelihood maximization) while sending failed samples to SDPO's targeted logit-level correction. The insight is that correct and incorrect samples carry distinct types of learning signal: correct samples should be reinforced, while incorrect samples should be analyzed and corrected. Treating both identically, as standard RL or standard self-distillation would, wastes the distinctive information each carries. SRPO raises the average by 3.4% over GRPO and 6.3% over SDPO, confirming that the optimal external-feedback strategy is not a single algorithm but a routing function that matches each sample to its most informative training objective. Semantic Soft Bootstrapping (SSB) (Mitra & Ulukus, 2025) offers a complementary in-context approach, providing the model with both correct and incorrect rollouts as exemplars that shape the next generation, achieving +10.6% on MATH-500 over GRPO.

The progression from SD-ZERO through RLSD, SDPO, and SRPO traces a clear arc: as external feedback grows richer and more targeted, the coupling between self-distillation and environmental grounding tightens. This very coupling, however, raises a fundamental question about the long-run stability of self-play without external anchors, a question that the saturation analysis below addresses.

Why Self-Play Saturates and Breakthrough Strategies

A critical theoretical bottleneck in self-distillation is the rapid onset of policy saturation. After a few iterations of SPIN (Chen et al., 2024) or OPSD (Zhao et al., 2026b), empirical observations universally note a plateau or severe degradation. This phenomenon is closely related to the "Ouroboros" problem (a model eating its own tail) and represents a distinct failure mode from exposure bias. Whereas exposure bias arises from train–test distribution mismatch in off-policy settings, saturation arises from the *absence* of distributional diversity in fully on-policy self-play.

Mathematically, this connects to mode collapse in GANs. Because the student optimizes against a target sharing its own inductive biases and architectural limitations, the hypothesis space progressively collapses. If the model discovers a syntactic "hack" or a highly confident but flawed reasoning heuristic, no external signal penalizes it. The model self-reinforces this flawed trajectory, driving $p_\theta(y_{flawed}|x) \to 1$. Once entirely certain of its own hallucinations, gradients vanish, exploration ceases, and the policy is trapped.

Strategies to break through saturation. Two primary approaches have emerged. First, *external verifiers and symbolic grounding*: as demonstrated in OPSD (Zhao et al., 2026b), self-play must be anchored by non-differentiable truth metrics (code execution engines, symbolic math verifiers). Even if the model becomes highly confident in a flawed derivation, the verifier assigns $R = 0$, shattering the self-reinforcing echo chamber. Second, *the distillation–RL loop*: to maintain trajectory diversity, state-of-the-art frameworks interleave self-distillation with Proximal Policy Optimization (PPO)-based RL:

$$\max_\theta \mathbb{E}_{y\sim p_\theta}\left[R(y) - \beta D_{\mathrm{KL}}(p_\theta \parallel p_{\mathrm{ref}})\right] \tag{30}$$

where $R(y)$ is provided by an independently trained reward model. Periodically refreshing the reward model using human-annotated preference data over the student's newest generations prevents premature convergence, ensuring that the self-play dynamic remains a continuously moving target.

A complementary failure mode is identified by Kim et al. (2026a), who trace self-distillation degradation in mathematical reasoning to the suppression of *epistemic verbalization*, the model's expression of uncertainty during reasoning. When self-distillation shortens reasoning traces (a common desirable outcome), it disproportionately removes hedging phrases and uncertainty markers that serve as implicit "attention flags" for difficult sub-problems. The result is shorter but less calibrated reasoning, where the model confidently commits to flawed steps it would have reconsidered in longer traces. This calibration degradation connects directly to the teacher uncertainty challenge in Section 6.3: just as an uncertain teacher produces unreliable distillation signals, an overconfident student produces unreliable self-play targets, and both pathologies share the same root cause in the suppression of distributional uncertainty.

6 Understanding OPD

The preceding two sections have surveyed a rich body of OPD methods, from teacher-guided approaches that leverage external expertise (Section 4) to self-distillation techniques that extract knowledge from within a single model (Section 5). These sections addressed the *how* of on-policy distillation. This section turns to the equally important *why* and *when*: under what conditions does OPD succeed, how does it fail, and what theoretical frameworks unify the diverse algorithmic choices surveyed above? Answering these questions is essential for moving OPD from an empirical art, where practitioners try multiple configurations and hope for the best, toward a principled engineering discipline with predictable outcomes.

Crucially, these four dimensions of understanding are not independent: they form a cumulative argument. Identifying *when* OPD works (Section 6.1) reveals the preconditions that the theoretical framework in Section 6.3 must explain. Cataloging *how* it fails (Section 6.2) motivates the specific design choices, adaptive divergences, curriculum weighting, and hybrid RL objectives, whose theoretical grounding Section 6.3 provides. Both the success conditions and the failure modes then inform the practical cost-benefit analysis in Section 6.4, which distils the theoretical insights into engineering guidelines. We organize this analysis accordingly: first preconditions, then failure modes, then the theoretical framework that explains both, and finally the practical tradeoffs that translate theory into deployment decisions.

6.1 Success Conditions

Understanding when OPD succeeds is as important as understanding how it works. Not all teacher-student pairs benefit equally from on-policy training, and identifying the preconditions for effective distillation can prevent wasted compute on configurations that are unlikely to yield meaningful gains. The success conditions identified here also set the terms for the failure-mode analysis that follows: failures are often best understood as violations of these preconditions.

Li et al. (2026d) provide the most systematic investigation to date of OPD training dynamics, identifying two necessary conditions for effective distillation. First, *thinking-pattern consistency*: the student and teacher must share compatible reasoning patterns, operationalized as a high overlap ratio in their top-k token distributions. Even when the teacher achieves higher benchmark scores, a mismatched thinking pattern (e.g., a non-thinking teacher distilling into a thinking student) causes low initial overlap that training cannot fully recover. Second, *genuine new knowledge*: the teacher must provide capabilities beyond what the student has already acquired. When both models are trained on the same data and recipe, they converge to similar distributions at their respective scales, leaving the teacher with little transferable signal, as validated through reverse distillation experiments showing that same-family 1.5B and 7B teachers are distributionally indistinguishable from the student's perspective.

At the token level, successful OPD is characterized by progressive alignment on high-probability overlap tokens, a small shared token set concentrating 97–99% of the probability mass. When these conditions fail, the authors propose two recovery strategies, off-policy cold start (SFT initialization that reduces the pattern gap) and teacher-aligned prompt selection (using the teacher's post-training prompts to sharpen alignment). Critically, they show that the quality of OPD's dense supervision degrades systematically with trajectory depth, with instability originating at later tokens and propagating backward, raising fundamental questions about OPD's applicability to long-horizon reasoning.

These findings point toward a general principle: OPD's benefit scales with the *exploitable gap* between teacher and student. When this gap is too small (same-recipe models), the teacher offers no new information. When this gap is too large (thinking-pattern mismatch), the student cannot absorb the teacher's supervision. The productive regime lies in between, and identifying this regime is the central diagnostic challenge that the failure-mode analysis in the next subsection addresses.

6.2 Failure Modes

Complementing the success conditions above, a growing body of work catalogs the specific ways in which on-policy distillation can fail. The success conditions identified in the previous subsection described structural preconditions: compatible thinking patterns and exploitable teacher-student gaps. The failure modes described here are mechanistic: they arise even when these preconditions hold, as artifacts of the on-policy training dynamics themselves. Together, these two subsections define the full diagnostic space for OPD practitioners, with success conditions governing *configuration choices* and failure modes governing *algorithmic choices* within a given configuration.

Failure modes of sampled-token OPD. Fu et al. (2026) provide a systematic empirical analysis of why the dominant sampled-token variant of OPD is fragile in long-horizon settings. They identify three failure modes: (1) an imbalanced one-token signal that reduces distribution matching to a single sample, (2) unreliable teacher guidance on student-generated prefixes that deviate from the teacher's training distribution, and (3) distortions caused by tokenizer or special-token mismatch between teacher and student. The second failure mode connects directly to the "flawed prefix trap" identified by SCOPE (Zheng et al., 2026) (Section 4.2): when the student generates a flawed reasoning step, the teacher's conditional distribution over the subsequent tokens is poorly calibrated, because the teacher has never been trained on such out-of-distribution prefixes. Matching this noisy teacher signal propagates errors rather than correcting them. Theoretically, token-level OPD is biased relative to sequence-level Reverse KL but enjoys a tighter worst-case variance bound, revealing a fundamental bias-variance tradeoff. Their proposed fix, teacher top-K local support matching via truncated Reverse KL with top-p rollout sampling and special-token masking, yields more stable optimization across math reasoning and multi-task agentic training. This remediation strategy is importantly different from the adaptive divergence approaches of Section 4.1: those methods address *which divergence to use* at each position, while this approach addresses *whether to trust the teacher* at all on a given prefix, revealing that the two dimensions of adaptation are complementary rather than redundant.

6.3 Unified Theoretical Perspectives

The success conditions and failure modes documented above raise a natural question: can they be explained by a single theoretical framework, or are they independent phenomena requiring separate analyses? The answer, as several recent works demonstrate, is that they share common mathematical roots. The success conditions (compatible thinking patterns, exploitable gaps) and the failure modes (flawed prefix trust, gradient anisotropy) both arise from the same underlying geometry of the divergence between teacher and student distributions. A unified theoretical perspective not only explains these phenomena but also reveals that many seemingly distinct methods are special cases of a common mathematical structure, reducing an overwhelming algorithmic landscape to a small number of interpretable design decisions.

The unified f-divergence perspective. As detailed in Section 4.1, the unified f-divergence framework of Wen et al. (2023) subsumes Forward KL, Reverse KL, JSD, and α-divergences as special cases parameterized by a single convexity function. This theoretical unification has a deeper implication: since all these divergences can be expressed as expectations under the student's on-policy distribution, the choice of divergence is a *regularization* decision rather than a fundamental algorithmic one. The generator function f determines the implicit weighting of likelihood ratios $p_T(y)/p_\theta(y)$, and different choices emphasize different regions of the ratio space. Forward KL up-weights regions where the student underestimates the teacher (coverage), while Reverse KL up-weights regions where the student overestimates (precision). This ratio-weighting interpretation explains why adaptive methods that switch between divergences based on local distributional geometry (Section 4.1) consistently outperform any fixed choice.

Geometric explanation for adaptive divergences. The empirical success of adaptive divergence selection (ToDi, AKL, Entropy-Aware OPD) admits a natural interpretation through information geometry. The Fisher information metric on the statistical manifold of autoregressive distributions varies across token positions. At reasoning-critical positions, the manifold is highly curved (the teacher's distribution concentrates on few tokens), making Reverse KL's mode-seeking behavior locally appropriate. At generation-flexible positions, the manifold is nearly flat (many synonyms are equally valid), and Forward KL's mode-covering behavior preserves this entropy. Under this view, adaptive methods implicitly navigate this varying curvature by selecting the divergence whose local geometry best matches the manifold structure at each position.

OPD as KL-constrained reinforcement learning. As detailed in Section 4.5, G-OPD (Yang et al., 2026c) proves that standard OPD is a special case of dense KL-constrained RL, establishing a formal equivalence between minimizing on-policy KL divergence and maximizing token-level rewards subject to a KL penalty. This equivalence bridges the distillation and RL communities, providing a shared analytical language for comparing methods that were previously studied in isolation. The practical consequence is that any advance in KL-constrained RL (better trust regions, adaptive penalty coefficients, variance reduction techniques) immediately transfers to OPD, and vice versa.

Unified KD+RL framework. The unified KD+RL framework of Li et al. (2025) (Section 4.5) provides a formal mathematical foundation for understanding why hybrid approaches consistently outperform either KD or RL alone. Their gradient decomposition, separating the dense KD gradient for token-level imitation from the Monte Carlo RL gradient for reward maximization, uncovers a complementary stabilization mechanism: the KD component dampens the high variance inherent in policy gradient estimation, while the RL component prevents the student from collapsing onto teacher modes that are suboptimal for the downstream task. This theoretical insight has practical implications for training schedules, as the optimal KD-to-RL weighting ratio should decrease over training as the student's policy stabilizes and can benefit more from reward-driven exploration.

Together, these theoretical perspectives converge on a central insight: on-policy distillation, reinforcement learning, and preference optimization are not separate training approaches but rather points on a continuous spectrum parameterized by the density of supervision, the choice of divergence, and the source of the training signal. This unification also resolves an apparent paradox in the failure-mode literature: different failure modes correspond to different regions of this spectrum where the chosen parameterization is locally suboptimal. Flawed-prefix trust failures arise when the divergence choice ignores teacher uncertainty (a problem the f-divergence framework addresses through adaptive weighting). Gradient anisotropy failures arise when the KD-to-RL balance is wrong (a problem the KD+RL equivalence addresses through reward scaling). With this unified perspective in hand, we can now approach the practical question of when to deploy OPD versus simpler off-policy pipelines with theoretical grounding rather than empirical heuristics.

6.4 On-Policy vs Off-Policy: Comparative Analysis and Cost

The theoretical frameworks above establish that on-policy training offers principled advantages through distributional alignment. But theory alone cannot settle the question of when practitioners should invest in on-policy infrastructure versus relying on simpler off-policy pipelines. This subsection examines the on-policy versus off-policy tradeoff through the lens of the most prominent off-policy success story, then analyzes the compute economics that govern practical deployment decisions.

DeepSeek-R1: Off-Policy Reasoning Distillation DeepSeek-R1 (DeepSeek-AI et al., 2025) represents the most prominent off-policy distillation success, training students from 1.5B to 70B on approximately 800K samples of long chain-of-thought reasoning traces from a 671B MoE teacher. The entire process is purely off-policy SFT, using no student rollouts during training.

Why Off-Policy Distillation Works So Well Here The effectiveness of this off-policy approach demands explanation, as it seemingly contradicts the core thesis of this survey that on-policy training mitigates exposure bias. Three factors account for this apparent paradox.

First, *data quality dominates data distribution*. The 800K samples from DeepSeek-R1 contain extremely high-quality reasoning chains with step-by-step verification, self-correction, and structured deliberation. When the teacher data is this rich and diverse, the exposure bias problem is partially alleviated because the static dataset already covers a wide range of reasoning patterns and error-recovery strategies.

Second, *the reasoning traces are inherently self-correcting*. Unlike typical off-policy data (e.g., teacher-generated summaries), R1's long CoT outputs contain backtracking, "wait, let me reconsider" moments, and multi-path exploration within a single sequence. The student learns not just the correct answer but the meta-cognitive process of verifying and correcting intermediate steps, which partially compensates for the lack of on-policy exploration.

Third, *the student's task is memorization of structure, not exploration of alternatives*. For mathematical reasoning, there are relatively few valid solution paths compared to open-ended generation. The off-policy data covers the dominant valid paths, and the student's primary challenge is faithfully reproducing these structured chains rather than discovering novel ones.

The Case for On-Policy Methods on Top of R1 Distillation However, the off-policy ceiling is real. The DeepSeek-R1 paper (DeepSeek-AI et al., 2025) explicitly notes that incorporating RL could further boost model performance but chose not to include it. Subsequent work has demonstrated that on-policy methods applied *after* off-policy R1-style distillation yield meaningful further gains.

RLKD (Xu et al., 2025b) demonstrates that structural reward signals can markedly improve sample efficiency in reasoning distillation, surpassing standard SFT-RL pipelines even on 0.1% of data. KDRL (Xu et al., 2025a) provides a unified framework that jointly optimizes KD and RL objectives during post-training, preventing the student from drifting too far from the teacher's policy while still benefiting from reward-driven exploration. RLAD (Zhang et al., 2026e) takes this unification further with a selective imitation strategy, where the student follows the teacher's guidance only when doing so improves the policy update. As discussed in Section 4.5, OPSD (Zhao et al., 2026b) and PACED (Xu et al., 2026a) represent the next evolution, applying genuine on-policy self-distillation to reasoning, addressing the residual exposure bias that off-policy R1 distillation cannot resolve.

Empirical Results: Off-Policy R1 Distillation The DeepSeek-R1 off-policy distillation results (DeepSeek-AI et al., 2025) provide a compelling performance baseline. On AIME 2024 (pass@1), R1-Distill-Qwen-1.5B scores 28.9%, already surpassing GPT-4o (9.3%) and Claude-3.5-Sonnet (16.0%). R1-Distill-Qwen-7B reaches 55.5%, R1-Distill-Qwen-14B achieves 69.7%, R1-Distill-Qwen-32B reaches 72.6%, and R1-Distill-Llama-70B achieves 70.0%. On MATH-500, the 7B distilled model reaches 92.8%, and the 32B model reaches 94.3%.

Distillation vs. Direct RL A further essential finding is that *off-policy distillation decisively outperforms direct RL for small-to-medium models*. At the 32B scale, applying GRPO directly to Qwen2.5-32B-Base achieves only 47.0% on AIME 2024, compared to R1-Distill-Qwen-32B at 72.6%, a gap of **25.6 percentage points**. This dramatic difference highlights that even off-policy distillation from a sufficiently capable teacher is far more effective than direct RL for smaller models.

Post-R1 Evolution: Qwen3 and Beyond Qwen3 (Yang et al., 2025a) refines the pipeline by introducing multi-stage training that combines off-policy distillation with subsequent on-policy RL. These developments suggest that the optimal reasoning distillation pipeline is a carefully staged hybrid: off-policy SFT on high-quality teacher data to establish a strong foundation, followed by on-policy RL or OPD to close the remaining distribution gap.

Why on-policy outperforms off-policy: theoretical grounding. DDT (Zhang et al., 2026b) provides formal theoretical justification for these empirical observations: the key factor driving RL's generalization advantage is on-policy data generation, which keeps training aligned with the model's evolving distribution. By showing that in-distribution finetuning and distribution re-alignment together bridge the SFT-RL generalization gap, DDT confirms that the on-policy training distribution itself is the primary mechanism.

Compute-Quality Tradeoff Analysis A neglected but critical consideration is the cost-benefit analysis of OPD. System designers constantly face the dilemma of allocating compute between more tokens (off-policy) versus better supervision (on-policy). We formulate the expected compute cost (in FLOPs) for N tokens as follows.

$$C_{\text{off}} \approx N \times (F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{31}$$

$$C_{\text{on}} \approx N \times (G_{\text{student}} + \lambda F_{\text{teacher}} + F_{\text{student}} + B_{\text{student}}) \tag{32}$$

where F, B denote forward and backward FLOPs, G is the autoregressive generation cost (scaling quadratically with sequence length due to KV cache updates), and $\lambda \in (0, 1]$ is the teacher supervision refresh rate. Because $G_{\text{student}} \gg F_{\text{student}}$, C_{on} is typically several times C_{off}.

Let $\mathcal{U}(\theta, C)$ represent the student's downstream utility given compute budget C. The constrained optimization is

$$\max \mathcal{U}(\theta, C) \quad \text{subject to } C = \gamma C_{\text{on}} + (1 - \gamma) C_{\text{off}} \leq C_{\max} \tag{33}$$

where γ is the fraction processed on-policy. For generic knowledge transfer and syntax alignment, $\gamma \to 0$ (off-policy) suffices. For reasoning, code generation, and complex mathematics, $\frac{\partial \mathcal{U}}{\partial C_{\text{on}}}$ vastly exceeds $\frac{\partial \mathcal{U}}{\partial C_{\text{off}}}$, justifying the compute multiplier. A natural hybrid curriculum is to begin with C_{off} to warm up latent representations, then progressively shift toward $\gamma \to 1$ in later training stages.

Concrete Cost Example To ground these formulas, consider distilling a 70B teacher into a 7B student on 8×H100 GPUs. Off-policy over 1B tokens, the teacher generates the dataset offline (∼200 GPU-hours), the student trains for ∼100 GPU-hours, totaling ∼300 GPU-hours. On-policy over 1B tokens, each step requires (1) student generation (∼3× forward-pass cost due to autoregressive decoding), (2) teacher scoring (one 70B forward pass), and (3) student backward pass, yielding ∼1,200–1,500 GPU-hours, a 4–5× overhead.

However, quality gains can be substantial. On instruction-following benchmarks, on-policy methods consistently outperform off-policy SFT by meaningful margins. For example, DistiLLM (Ko et al., 2024) achieves state-of-the-art ROUGE-L scores with notable training speedup over previous on-policy baselines. For reasoning tasks, the off-policy ceiling, i.e., the maximum achievable performance regardless of data volume, is strictly lower because static datasets cannot cover the combinatorial space of valid reasoning paths. This "ceiling gap" widens with task complexity, providing the strongest motivation for on-policy methods despite their higher cost.

Speculative KD (Xu et al., 2025c) partially closes this cost gap by leveraging the student's generations as candidate tokens that the teacher filters and selectively replaces (see Section 7.2 for a detailed treatment).

GPU Memory Overhead and Practical Solutions. Beyond FLOPs, the most severe constraint for white-box OPD is GPU memory. Holding a 70B teacher alongside a 7B student requires the teacher's weights (∼140 GB in BF16), the student's weights and optimizer states (∼84 GB), and, critically, the KV cache for generation and the full-vocabulary logits tensor ($[B, T, |V|]$) for distillation. Peak memory can easily exceed an 8×80 GB H100 node. Practitioners alleviate this via (1) *Teacher Quantization*, serving the teacher in FP8 or INT4 for a 2–4× memory reduction at negligible precision loss, (2) *Logit Offloading/Recomputation*, immediately offloading logits to CPU RAM or retaining only the top-k, and (3) *Aggressive Gradient Checkpointing*, minimizing student activation memory during the backward pass. Combining these techniques is essential for viable white-box OPD on standard clusters.

7 Applications and Systems

The preceding sections have developed OPD from foundational theory (Section 2), through teacher-guided (Section 4) and self-distillation (Section 5) methods, to a unified theoretical understanding (Section 6). This section examines how these algorithmic advances translate into practice. As OPD moves from academic prototypes to industrial-scale deployments, new challenges emerge that are invisible in controlled experimental settings: multi-teacher coordination, system-level throughput bottlenecks, cross-modal transfer, and privacy constraints. We organize these practical considerations into three areas: industrial deployment at foundation-model scale (Section 7.1), system-level optimizations that make OPD computationally viable (Section 7.2), and emerging domains that extend OPD beyond text-only language modeling (Section 7.3).

7.1 Industrial Deployment

The strongest validation of any training methodology is its adoption in production systems. Recent foundation models have increasingly incorporated OPD into their post-training pipelines, revealing three distinct industrial use patterns that align with the theoretical progression surveyed above.

Two-phase distillation pipelines. The most common industrial pattern combines off-policy cold-start with on-policy refinement. The Qwen3 technical report (Yang et al., 2025a) exemplifies this approach, employing strong-to-weak distillation where larger models provide logit-level supervision on student-generated on-policy sequences, with the student generating responses in either thinking or non-thinking mode. Gemma 2 (Gemma Team et al., 2024) similarly embeds knowledge distillation directly into pre-training to bridge performance gaps between parameter tiers. At larger scale, MiMo-V2 (Xiaomi LLM-Core Team et al., 2026) combines distillation from domain-specialized teachers with RL-based training, demonstrating that the boundaries among logit-based, outcome-based, and self-play feedback routes increasingly blur as systems mature.

OPD as model consolidation. A second industrial pattern uses on-policy distillation not for teacher–student compression but for *merging* multiple specialized experts into a single deployable model. KAT-Coder-V2 (Li et al., 2026a) decomposes agentic coding into five expert domains, each undergoing independent SFT and RL training, and then consolidates them via on-policy distillation, achieving 79.6% on SWE-bench Verified. Nemotron-Cascade 2 (Yang et al., 2026d) applies the same consolidation principle to a 30B Mixture-of-Experts model with only 3B activated parameters, achieving Gold Medal-level performance on IMO, IOI, and ICPC World Finals with 20$\times$ fewer parameters than DeepSeek-V3.2-Speciale.

OPD for controllable multi-budget reasoning. A third emerging pattern uses on-policy distillation not to consolidate *different tasks* but to fuse *different reasoning modes* of the same task into a single controllable model. ORBIT (Liang et al., 2026) addresses the practical deployment challenge that users need variable reasoning depth depending on query difficulty, but maintaining separate models for each budget level is operationally prohibitive. The framework operates in two phases. First, an expansion-compression loop traverses the accuracy–length Pareto frontier via RLVR under progressively tighter context budgets ($L_{k+1} = L_k/2$), producing stage-wise per-mode expert policies. Second, multi-teacher on-policy distillation fuses these experts into a single model through mode-aware Reverse KL:

$$\mathcal{L}_{\text{fusion}}(\phi) = \mathbb{E}_{k\sim U(1,K)} \mathbb{E}_{\substack{q\sim\mathcal{D} \\ o\sim\pi_\phi(\cdot|q,p_k)}} \left[\sum_t \log \frac{\pi_\phi(o_t|q, p_k, o_{<t})}{\pi_{\theta_k}(o_t|q, p_k, o_{<t})} \right] \tag{34}$$

where p_k is the mode-specific prompt and $\{\pi_{\theta_k}\}$ are the frontier experts. Model merging initializes the student to mitigate the cold-start problem identified by Jang et al. (2026): without initialization near the expert manifold, the student's on-policy rollouts rarely reach the teachers' high-density regions, starving the distillation objective of useful signal. Across DeepSeek-Distill-Qwen-1.5B, Qwen3-4B, and Openmath-Nemotron-7B, ORBIT produces models that smoothly interpolate between four reasoning modes (2K to 32K token budgets) with competitive performance at each budget level, matching the initial model's accuracy at the highest budget while providing 2–7$\times$ token savings at lower budgets. The connection to the consolidation pattern above is instructive: where KAT-Coder-V2 and Nemotron-Cascade 2 merge experts specialized by *domain*, ORBIT merges experts specialized by *compute budget*, suggesting that on-policy distillation is emerging as a general-purpose multi-teacher fusion tool for any axis of model specialization.

These three patterns reflect a broader shift in how industry views OPD. The first pattern treats it as a *quality refinement* tool (making an existing pipeline produce better outputs), the second treats it as an *architectural simplification* tool (collapsing a complex multi-expert system into a single deployable model), and the third treats it as an *inference-time control* mechanism (enabling a single model to serve multiple latency-accuracy operating points). The convergence of all three patterns toward on-policy generation, rather than static dataset distillation, confirms that the theoretical advantages of distributional alignment translate directly into industrial value.

7.2 System-Level Optimization

The algorithmic innovations above assume unlimited compute and perfect infrastructure. In practice, the dominant bottleneck of OPD is not the choice of divergence or curriculum but the sheer cost of the teacher forward pass on dynamically generated student tokens. This creates a *chicken-and-egg* problem: the student must generate sequences before the teacher can score them, but both steps are expensive and sequential. Addressing this bottleneck requires rethinking the interaction pattern between teacher and student at the systems level.

Speculative KD (Xu et al., 2025c) resolves the sequential dependency by drawing inspiration from speculative decoding: the student generates candidate token blocks, and the teacher evaluates each token against its own distribution using acceptance criteria. Tokens that the teacher is unlikely to produce are rejected and re-sampled from the teacher's distribution, yielding training sequences that combine on-policy exploration with teacher-quality corrections at a lower cost than full teacher generation. DistillSpec (Zhou et al., 2023) inverts this relationship: instead of using speculation to speed up training, it uses on-policy KD to improve the draft model for speculative decoding at *inference* time. The expected speedup follows from the acceptance rate:

$$E[S] = \frac{\mathbb{E}[K_{\text{accepted}}] + 1}{1 + c} \quad \text{where } \mathbb{E}[K_{\text{accepted}}] = \mathbb{E}_{y \sim P_S}\left[\sum_{k=0}^{K-1} \prod_{i=0}^{k} \min\left(1, \frac{P_T(y_i)}{P_S(y_i)}\right)\right] \tag{35}$$

where c is the ratio of student-to-teacher forward-pass cost. The connection between these two methods reveals a virtuous cycle: better distillation produces better draft models, which enable faster speculative decoding, which in turn enables more efficient distillation. Together with cross-tokenizer distillation (Boizard et al., 2025) that enables OPD across disparate model families, these systems innovations are steadily reducing the practical overhead that has historically limited OPD adoption.

7.3 Emerging Domains

The methods surveyed so far have focused predominantly on text-only language models, yet the core principles of OPD, training on the student's own distribution to eliminate exposure bias, are modality-agnostic. A rapidly growing body of work extends on-policy distillation to multimodal, agentic, and privacy-sensitive settings, each of which introduces domain-specific challenges that require adapting the foundational OPD framework.

Multimodal distillation. Extending OPD across modality boundaries introduces a core alignment challenge: the teacher and student may process entirely different input representations (text vs. images, text vs. audio), so standard logit matching requires an additional cross-modal bridging step. The dominant strategy is *modality-asymmetric self-distillation*, where the same model's text-conditioned version serves as a teacher for its perceptually-conditioned student. VOLD (Bousselham et al., 2025) demonstrates this for vision-language models, with its key finding that an SFT cold-start phase is essential for initial cross-modal alignment before on-policy training can begin. CORD (Hu et al., 2026) applies the same principle to audio, using the text-conditioned LALM as an internal teacher that supervises audio-conditioned rollouts through importance-aware weighted Reverse KL at the token level and GRPO at the sequence level, substantially bridging the audio–text reasoning gap with only 80K synthetic samples. X-OPD (Cao et al., 2026) extends this to speech by distilling from a text-mode LLM to a speech-mode LLM on student-generated speech inputs.

A second strategy applies teacher-guided OPD across modalities, where the teacher and student are genuinely different models rather than different modes of the same model. This strategy faces a harder alignment problem but enables compression across capability tiers. Video-OPD (Li et al., 2026c) extends on-policy distillation to temporal video grounding, where the sequential nature of video understanding creates exposure bias analogous to text generation, achieving state-of-the-art results while reducing training cost compared to GRPO. VLA-OPD (Zhong et al., 2026) brings OPD to robotic manipulation, where the action space is continuous rather than discrete, requiring the student to learn precise motor control from the teacher's dense token-level supervision on self-generated action trajectories, significantly improving sample efficiency over RL and robustness over SFT. At industrial scale, HY-Embodied-0.5 (Tencent Robotics X and HY Vision Team, 2026) validates this cross-modal teacher-guided approach by transferring capabilities from a 32B expert to a 2B embodied VLM, outperforming similarly sized models on 16 out of 22 benchmarks. The consistent finding across these domains is that on-policy generation is even more critical in multimodal settings than in text-only ones, because the diversity of valid outputs (multiple correct grasping trajectories, multiple valid temporal localizations) makes off-policy data coverage impractical.

Agentic distillation. While multimodal OPD extends across modality boundaries, agentic distillation extends across *temporal* boundaries, requiring the student to learn not just what to generate but when to act, which tools to invoke, and how to recover from environmental failures across multi-step interaction trajectories. The core challenge is that teacher supervision degrades with trajectory depth (as documented in Section 6.1), making naive application of single-turn OPD techniques insufficient.

Two complementary strategies address this challenge. The first is *targeted correction*: rather than providing complete demonstrations, the teacher corrects only the earliest error in the student's multi-step trajectory. SCoRe (Lyu et al., 2025) pioneered this approach, and OEL (Ye et al., 2026a) extends it to continuous post-deployment learning where interaction outcomes serve as dynamic correction signals. The second is *hindsight-guided supervision*: OpenClaw-RL (Wang et al., 2026b) observes that every agent interaction produces a next-state signal encoding two recoverable forms of supervision, evaluative signals (scalar rewards via PRM) and directive signals (textual hints from the next state that construct an enhanced teacher context). On Qwen3-4B across five environment types, combining both signal types consistently outperforms either alone.

Domain-specific instantiations further validate the agentic OPD paradigm and reveal a progression in the type of privileged information exploited. At the simplest level, Afsharrad et al. (2026) apply GKD to autonomous vehicle motion planning, where the teacher's privileged information is simply its larger capacity (Qwen3-8B), distilling into a 1.7B student with $5\times$ compression while substantially outperforming a dense-feedback RL baseline. Skill-SD (Wang et al., 2026a) goes further by *constructing* dynamic privileged information from the agent's own experience: completed trajectories are summarized into compact natural-language "skills" that describe successful behaviors and common mistakes, creating a self-improving PI signal that adapts as the agent encounters new task variants (+14.0% on AppWorld, +10.9% on Sokoban over GRPO). Privileged Information Distillation (Penaloza et al., 2026) provides theoretical grounding for why these approaches work, formalizing the conditions under which training-time PI is particularly effective and showing that the benefit of PI grows with task horizon length, which explains why agentic settings benefit disproportionately from PI-based self-distillation.

Privacy-preserving distillation. A distinct but increasingly important application domain is the intersection of OPD with differential privacy. DP-OPD (Khadem et al., 2026) provides the first principled framework for combining on-policy distillation with differential privacy, applying DP-SGD noise only to the student's gradient updates while keeping the teacher frozen and non-private. Under a strict privacy budget ($\varepsilon = 2.0$), DP-OPD achieves better perplexity than both DP fine-tuning and off-policy DP distillation on Yelp and BigPatent. We discuss the broader implications and open questions of privacy-preserving OPD in Section 8.

8 Open Problems and Future Directions

Despite the rapid progress surveyed in the preceding sections, several foundational questions remain unanswered and numerous practical bottlenecks persist. This section identifies the most promising research frontiers. Crucially, these directions are not independent wish-list items but form a logically connected agenda: the most immediate engineering challenge, scaling laws, determines when OPD is worth investing in at all. Once the investment decision is made, the quality of the teacher signal, addressed by the uncertainty-aware distillation direction, and the quality of the training curriculum, addressed by dynamic curriculum design, jointly determine the ceiling of what OPD can achieve. The architectural constraint of vocabulary mismatch then limits cross-architecture applicability, while the temporal and modality extension directions (agentic and multimodal OPD) expand the domains where these methods can be deployed. Privacy-preserving distillation and the distillation-RL loop closure complete the agenda by addressing deployment constraints and the fundamental training-time integration challenge. We present these directions roughly in this logical order, from immediate bottlenecks to longer-term scientific questions.

Distillation Scaling Laws. Chinchilla scaling laws (Hoffmann et al., 2022) provide principled compute allocation for pre-training, and Busbridge et al. (2025) recently extended this framework to off-policy knowledge distillation, fitting parametric scaling curves that reveal optimal teacher size grows sub-linearly with compute budget. However, no equivalent scaling law exists for on-policy distillation, where the generation cost of student rollouts introduces an additional compute axis that reshapes the optimization problem. Practitioners currently rely on costly trial-and-error to determine how much on-policy generation budget to allocate relative to student and teacher size. A natural conjecture is that the distillation loss follows a joint power-law of the form

$$L(N_S, N_T, D_{\text{on}}) = E + \frac{A}{N_S^{\alpha}} + \frac{B}{N_T^{\beta}} + \frac{C}{D_{\text{on}}^{\gamma}} + f(N_S, N_T) \tag{36}$$

where E is the irreducible task entropy and $f(N_S, N_T)$ models the capacity-gap interference between student and teacher. While no study has yet validated this functional form, emerging empirical evidence hints at its plausibility. Wu et al. (2026) provide indirect evidence through their teacher consistency analysis: the gradient discrepancy between online and offline OPD is bounded by the χ^2 divergence between the student's current and reference policies, suggesting that the on-policy generation term D_{on} interacts non-trivially with the training trajectory rather than contributing independently.

DeepSeek-R1 (DeepSeek-AI et al., 2025) provides further empirical grounding. Its results illuminate how student capacity affects distillation quality: on AIME 2024, performance scales as 28.9% $\to$ 55.5% $\to$ 69.7% $\to$ 72.6% for student sizes 1.5B $\to$ 7B $\to$ 14B $\to$ 32B. The steepest gain occurs between 1.5B and 7B (26.6% absolute), while the 14B$\to$32B jump yields only 2.9%, suggesting that α in Equation 36 may be small (rapid diminishing returns with student scale) while the teacher capacity term B/N_T^{β} dominates. The Qwen3 experience (Yang et al., 2025a) further suggests that $f(N_S, N_T)$ is non-trivial, as distilling from multiple specialized teachers yields better scaling than from a single monolithic teacher of equivalent total parameters.

Future work should conduct controlled grid searches that independently vary N_S, N_T, and D_{on} to disentangle these effects. Such studies would enable practitioners to answer fundamental allocation questions: "Given a fixed GPU budget, should one distill from a 70B teacher for 1B tokens or a 405B teacher for 200M tokens?" The precision-recall tradeoff framework of Cha & Cho (2025) provides a useful lens for such investigations. That work demonstrates that distillation induces a fundamental trade-off between sample precision and distributional coverage, modulated by a single entropy-controlling parameter. On-policy generation may shift this trade-off by allowing the student to explore its own distribution, potentially improving recall without sacrificing precision. Establishing such scaling laws would transform OPD from an empirical art into a principled engineering discipline analogous to Chinchilla-guided pre-training. However, even with optimal resource allocation, the quality of distillation is bounded by the reliability of the teacher's supervisory signal, which brings us to the second open challenge.

Teacher Calibration and Uncertainty-Aware Distillation. White-box OPD assumes that the teacher's logit distribution provides a reliable, high-quality supervisory signal at every token position. In practice, this assumption frequently breaks down. Teacher models suffer from overconfidence, hallucination, and poor calibration (Jin et al., 2026), especially when evaluated on out-of-distribution prefixes generated by an exploring student. When the student produces a flawed reasoning step, the teacher may assign high probability to a hallucinated continuation, pulling the student into an error cascade where minimizing KL against an uncertain teacher forces the student to confidently mimic noise.

Several existing methods partially address this concern. Entropy-Aware OPD (Jin et al., 2026) gates the divergence choice based on teacher entropy, switching from Reverse KL to Forward KL when the teacher is uncertain. GATES (Stein et al., 2026) uses consensus-based gating to suppress the distillation signal when the tutor demonstrates low agreement across multiple samples. PACED (Xu et al., 2026a) proves that the gradient signal-to-noise ratio vanishes when the teacher-student gap is too large, implicitly addressing calibration through curriculum design. SCOPE (Zheng et al., 2026) uses teacher perplexity to weight the distillation loss on incorrect rollouts, down-weighting instances where the teacher is uncertain about the student's flawed prefix. Most directly, Li et al. (2026d) show that even when a failing teacher provides globally informative reward (AUROC $\sim$0.75), the per-token advantages become anisotropic across positions within each sequence, creating a locally flat reward landscape that renders token-level gradients ineffective.

However, none of these methods explicitly model the teacher's epistemic versus aleatoric uncertainty. A promising direction is to decompose teacher uncertainty into its epistemic component (which should attenuate the distillation loss) and its aleatoric component (which reflects genuine task ambiguity and should be preserved in the student's output distribution). Achieving this decomposition efficiently, without requiring expensive Bayesian inference over the teacher's parameters, remains an important open challenge. Even a reliable teacher signal, however, is wasted if the training curriculum presents problems that are too easy or too hard for the student's current capacity, which motivates the next direction.

Dynamic Curriculum Distillation. Uniformly sampling prompts during OPD ignores the student's evolving capacity. Exposing a novice student to complex reasoning prompts yields gradient noise, while exposing an advanced student to trivial prompts wastes compute. PACED (Xu et al., 2026a) demonstrates that a Beta-kernel weighting scheme, which assigns maximum weight to sequences near the student's competence frontier, dramatically improves sample efficiency and is provably minimax-robust. AdaSwitch (Peng et al., 2025) introduces adaptive switching at the token level, selectively integrating teacher guidance only when the student's divergence exceeds a context-aware threshold. Lion (Jiang et al., 2023) implements a teacher-driven curriculum that progressively generates harder instructions targeting the student's identified weaknesses.

Despite these advances, principled curriculum design for OPD remains largely ad hoc. The core challenge is that the optimal training distribution shifts continuously as the student improves, requiring dynamic adaptation rather than fixed schedules. A promising approach is to draw on the "zone of proximal development" concept formalized in PACED (Xu et al., 2026a): at each training step, the student should train on prompts at the boundary of its current capabilities, where the solve rate is neither too high (wasting compute) nor too low (producing uninformative gradients). Another underexplored direction is adapting offline data curation techniques to on-policy settings. MiniPLM (Gu et al., 2025) demonstrates that selecting training instances based on the log-probability discrepancy between teacher and a reference model (Difference Sampling) substantially improves pre-training efficiency under off-policy KD. Transplanting this principle to on-policy rollouts, using teacher-reference divergence to prioritize which student-generated sequences deserve dense teacher feedback, could yield significant compute savings by avoiding teacher inference on uninformative rollouts. Scaling these principles to large prompt pools, including automatically estimating prompt difficulty, tracking the student's evolving competence boundary, and efficiently allocating compute across difficulty tiers, remains ripe for exploration. The interaction between curriculum scheduling and divergence adaptation (e.g., using Forward KL early for broad coverage, then switching to Reverse KL for consolidation) further expands the design space. Effective curriculum design in turn requires that the student and teacher occupy commensurable representation spaces, a prerequisite violated when teacher and student come from different model families, which the next direction addresses.

Latent Space Distillation for Vocabulary Mismatch. Most OPD objectives assume that teacher and student share the same vocabulary, an assumption increasingly violated in practice when distilling across model families (e.g., Llama into Qwen). Marginalizing over misaligned tokenizers is computationally prohibitive and destroys high-frequency semantic signals. DSKD (Zhang et al., 2025b) addresses this through dual-space projectors that map hidden states between teacher and student representation spaces, while cross-tokenizer KD (Boizard et al., 2025) uses optimal transport to align logit distributions across vocabularies.

These methods operate at the output logit level, where the vocabulary bottleneck has already compressed the teacher's internal representations. A more ambitious direction would bypass the vocabulary layer entirely and distill in latent space, matching the geometric structure of the teacher's hidden-state manifold rather than its token-level predictions. This could enable seamless cross-architecture OPD where teacher and student differ not only in vocabulary but also in hidden dimension, number of layers, and attention structure. Early steps in this direction include DSKD's dual-space formulation (Zhang et al., 2025b), but extending latent distillation to on-policy settings (where the student's rollouts produce hidden states that must be aligned with the teacher's counterfactual representations) introduces additional challenges around representation drift during training. Resolving the vocabulary mismatch challenge is a prerequisite for extending OPD to its most demanding deployment context, autonomous agents, where the action space involves diverse tool-call schemas that compound the vocabulary heterogeneity problem.

On-Policy Distillation for Autonomous Agents. Current OPD methods overwhelmingly target single-turn generation or linear chains of thought, yet LLMs are increasingly deployed as autonomous agents interacting with external environments such as APIs, terminals, databases, and web browsers. In agentic settings, the action space is vast, feedback is highly delayed, and token-level distillation fails to capture the long-horizon credit assignment required for multi-step planning.

As surveyed in Section 7.3, initial efforts such as SCoRe (Lyu et al., 2025), OEL (Ye et al., 2026a), Privileged Information Distillation (Penaloza et al., 2026), OpenClaw-RL (Wang et al., 2026b), Afsharrad et al. (2026), and Skill-SD (Wang et al., 2026a) provide early realizations of agent-level distillation. However, Li et al. (2026d) provide direct evidence that OPD's dense supervision degrades systematically with trajectory depth: teacher accuracy advantage decreases monotonically as the prefix grows longer, and training instability originates at later tokens before propagating backward through the sequence. This back-to-front collapse pattern suggests that extending OPD to agentic settings, where trajectories span hundreds of tool calls over minutes or hours, could require qualitatively new supervision mechanisms rather than straightforward scaling of existing token-level objectives.

Several fundamental challenges remain open beyond this depth-degradation issue. First, *environment non-stationarity*: unlike text generation where the "environment" (the prompt) is fixed, agentic environments change in response to actions, making direct trajectory comparison between teacher and student meaningful only when the teacher provides counterfactual evaluations conditioned on the student's actual state. Second, *tool-use combinatorics*: modern agent frameworks expose dozens of tools with structured argument schemas, suggesting that distillation may need to operate at the tool-call level rather than individual tokens. Third, *safety-critical credit assignment*: in coding agents, a single incorrect file write can corrupt a repository, and in web-browsing agents, a misguided click can trigger irreversible actions. Future distillation frameworks should incorporate safety constraints that prevent the student from exploring catastrophically dangerous action sequences during on-policy generation, even when those sequences might be maximally informative for learning. Developing OPD methods that balance exploration with safety in agentic settings remains a largely uncharted territory. Closely related to the temporal-extension challenge is the modality-extension challenge: just as agentic OPD must extend supervision across time, multimodal OPD must extend it across modality boundaries, where both dimensions interact in robot manipulation and embodied agent settings.

Multimodal On-Policy Distillation. While the preceding direction extends OPD across temporal horizons, an equally important frontier extends it across modality boundaries. Existing OPD methods overwhelmingly target text-only LLMs, but multimodal models present distinct challenges. Vision-language models face a unique difficulty, as high-quality visual reasoning data is scarce while text-based reasoning data is abundant and easily verifiable.

The initial results surveyed in Section 7.3 are encouraging, but several questions remain. How should the distillation objective account for information asymmetry between modalities (e.g., when the teacher processes text but the student processes images of the same content)? Can on-policy distillation transfer spatial reasoning, temporal grounding, and audio understanding simultaneously, or does each modality require specialized objectives? The interaction between modality-specific data scarcity and on-policy data generation, where the student explores its own multimodal output space, is particularly promising, as OPD's self-correcting dynamics could partially compensate for the lack of large-scale annotated multimodal reasoning data. Both the agentic and multimodal frontiers involve deploying OPD in regulated or sensitive environments, which requires confronting a dimension that the text-only research community has largely deferred: privacy.

Privacy-Preserving Distillation. As LLMs are increasingly adapted to proprietary and domain-specific corpora containing sensitive information, the intersection of on-policy distillation and differential privacy (DP) is emerging as a critical research direction. As discussed in Section 7.3, DP-OPD (Khadem et al., 2026) provides the first principled framework for this setting, with an asymmetric design that applies DP-SGD only to the student while keeping the teacher frozen and non-private. This asymmetry is a key innovation: prior approaches either apply DP-SGD to both teacher and student (doubling compute and worsening the privacy–utility tradeoff) or rely on DP synthetic text generation from a DP-trained teacher (requiring DP-optimization of the large teacher itself).

The on-policy formulation is particularly well-suited to the DP setting because the student trains on its own generated prefixes, exactly the distribution it will encounter at inference time, thereby reducing the compounding errors that DP noise amplifies in autoregressive generation. Open questions include scaling DP-OPD to modern reasoning-capable LLMs where longer rollouts increase the privacy cost per step, and integrating DP with self-distillation methods where no separate teacher exists. The privacy constraints explored here also impose structure on when distillation can alternate with reinforcement learning, which connects to the final and most algorithmically fundamental open challenge.

Closing the Distillation-RL Loop. Distillation and reinforcement learning are historically treated as separate, linearly staged processes, either distill then RL, or RL then distill. However, static distillation saturates when the student exhausts the teacher's knowledge, and pure RL diverges when the reward signal is sparse. The true performance ceiling likely lies in closing the loop by alternating between KD-driven stabilization and RL-driven exploration.

Empirical evidence supports this view. Chen et al. (2025) show that across Llama and Qwen families, RL-based post-training retains more pre-training capabilities than SFT, precisely because RL generates on-policy rollouts that maintain distributional proximity to the student's prior, suggesting that the on-policy training distribution itself, not the reward signal per se, is the primary mechanism through which RL avoids forgetting. SDFT (Shenfeld et al., 2026) demonstrates that self-distillation can serve as a principled RL alternative for continual learning, achieving comparable forgetting resistance without requiring explicit reward functions. KDRL (Xu et al., 2025a) jointly optimizes KD and RL objectives during post-training, preventing the catastrophic forgetting that plagues sequential pipelines. The unified framework of Li et al. (2025) formally subsumes both KD and RL as special cases, with the dense KD gradient stabilizing early training and the RL gradient enabling exploration beyond the teacher.

Despite these advances, principled methods for alternating between distillation and RL remain open challenges. Determining when to switch objectives, how to prevent mode collapse during RL phases, and how to ensure that RL-discovered capabilities are not overwritten during subsequent distillation phases all remain unresolved. Closing this loop would also substantially change the evaluation question: if a model alternates between distillation and RL throughout its lifecycle, what counts as a valid benchmark for its final capability, and how should we evaluate robustness to distribution shift rather than static pattern matching?

Evaluation Methodology Beyond Benchmarks. Standard benchmarks (e.g., MMLU, GSM8K) suffer from data contamination and measure only static pattern matching, failing to capture the intrinsic robustness, out-of-distribution generalization, and hallucination rates of a distilled student relative to its teacher. Fu et al. (2026) systematically demonstrate that token-level OPD can appear successful on standard benchmarks while failing catastrophically on distribution-shifted prompts, highlighting the gap between benchmark performance and true generalization.

Future evaluation methodology should move toward dynamic, adversarial testing that probes whether the student has learned the underlying causal mechanism of reasoning rather than superficial correlations. Promising directions include systematically testing on semantically equivalent but syntactically diverse prompts, measuring the student's calibration under distribution shift, and assessing whether distilled models preserve the teacher's uncertainty structure.

Practical Guidelines and Decision Framework. The sheer volume of OPD techniques surveyed in this paper, spanning off-policy baselines, token-level and sequence-level methods, reward-guided approaches, and self-distillation, can cause decision paralysis for engineering teams. Based on the comparative analysis across Sections 4–7, we distill several actionable guidelines.

When to use off-policy distillation. For general instruction-following and chat, off-policy SFT on teacher-generated data remains the most cost-effective starting point. High-quality teacher data combined with preference optimization (Zhang et al., 2025a) can achieve competitive performance at a fraction of the on-policy compute cost. Start here and measure the gap before investing in on-policy infrastructure.

When on-policy becomes necessary. The transition to on-policy distillation is justified when (1) the student exhibits clear exposure bias, performing well on teacher-style prompts but failing on user-generated ones with different phrasing, (2) the task involves multi-step reasoning where compounding errors degrade quality (Ross et al., 2011; Gudibande et al., 2023), or (3) the student needs to exceed the teacher in a specific domain via reward-guided exploration (Yang et al., 2026c).

Choosing between logit-level and reward-level OPD. If the teacher is white-box and the student is small (<7B), logit-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) provide the dense supervision small models need. For larger students ($\geq$7B) with outcome verifiers, reward-guided OPD (RLKD (Xu et al., 2025b), AlignDistil (Zhang et al., 2025a)) enables discovery of novel solution paths. The hybrid approach, combining off-policy SFT warm-up (as in DeepSeek-R1 (DeepSeek-AI et al., 2025)) with on-policy logit-level or reward-guided fine-tuning, consistently yields the best results across the surveyed literature.

The "distillation tax" budget rule. Based on the compute analysis in Section 6.4 and the staged pipelines described in recent industrial reports (DeepSeek-AI et al., 2025; Yang et al., 2025a), a practical rule of thumb is to allocate the majority of the training budget to off-policy warm-up, reserve a smaller fraction for on-policy logit distillation, and dedicate the final phase to reward-guided refinement. This staged approach front-loads cheap, high-bandwidth learning and reserves expensive on-policy compute for the final quality push.

9 Conclusion

This survey has systematically examined on-policy distillation for large language models, synthesizing representative works that span foundational formulations, algorithmic innovations, theoretical analyses, and industrial deployments.

Key Findings. Three central insights emerge from this synthesis. *First*, shifting from off-policy to on-policy training distributions, from $y \sim p_{\text{data}}$ to $y \sim p_\theta$ (Section 2), is a consistently impactful design choice, yielding meaningful accuracy gains across mathematical reasoning, code generation, and instruction following (Tables 2 and 3). The improvement stems from eliminating exposure bias: the student learns to recover from its own errors rather than memorizing idealized trajectories. *Second*, divergence selection is not one-size-fits-all but must be adapted to both task and token position. As Section 4.1 shows, Reverse KL excels at reasoning tasks where mode-seeking prevents hallucination, while Forward KL preserves the diversity needed for open-ended generation. Adaptive methods such as ToDi (Jung et al., 2025) and Entropy-Aware OPD (Jin et al., 2026) that dynamically select divergences represent the current frontier. *Third*, the boundary between distillation and reinforcement learning is rapidly dissolving. G-OPD (Yang et al., 2026c), RLAD (Zhang et al., 2026e), and the hybrid RL+IL framework of Li et al. (2025) show that the optimal training signal combines dense teacher supervision with sparse outcome rewards, with KD providing stability while RL enables exploration beyond the teacher.

Practical Takeaways. Our survey distills several actionable guidelines (Section 8). With white-box teacher access, token-level methods (GKD (Agarwal et al., 2024), DistiLLM (Ko et al., 2024)) offer the best quality-compute tradeoff for general instruction following. For reasoning-intensive tasks, sequence-level methods (MiniLLM (Gu et al., 2024)) or hybrid approaches (Fast OPD (Zhang et al., 2026a), PACED (Xu et al., 2026a), Lightning OPD (Wu et al., 2026)) are preferable despite higher variance. Lightning OPD in particular demonstrates that offline distillation under teacher consistency can match online OPD at $4\times$ lower cost, significantly lowering the barrier for academic research. With only API access, GAD (Ye et al., 2025) enables effective on-policy learning through adversarial formulations. Self-distillation methods (SPIN (Chen et al., 2024), OPSD (Zhao et al., 2026b), SDPO (Hübotter et al., 2026), SD-ZERO (He et al., 2026)) eliminate the need for a separate teacher entirely, making them attractive under resource constraints. SD-ZERO further shows that binary verification rewards can be converted into dense self-supervision through self-revision. The compute overhead of on-policy generation, typically 3–8$\times$ that of off-policy training, can be substantially reduced via speculative decoding (Xu et al., 2025c), prefix truncation (Zhang et al., 2026a), or offline precomputation (Wu et al., 2026).

Looking Ahead. The most pressing open problems are threefold. First, the absence of distillation scaling laws analogous to Chinchilla (Hoffmann et al., 2022) forces practitioners to rely on costly trial-and-error for compute allocation. Second, the field needs uncertainty-aware objectives that prevent the "echo chamber" effect when students explore out-of-distribution regions, where Li et al. (2026d) show that even globally informative teacher reward can be locally unexploitable due to anisotropic per-token advantages. Third, extending OPD to multi-turn agentic settings, where training involves entire interaction trajectories rather than single completions, remains a fundamental challenge compounded by the systematic degradation of teacher supervision with trajectory depth (Li et al., 2026d). As the field matures, we anticipate convergence toward unified KD+RL frameworks that treat distillation not as a separate training stage but as a continuous regularization mechanism throughout the model's lifecycle, from pre-training through deployment.

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